

Software Requirements Specification (SRS)

EdTech Study Platform – Smart-Gated Advanced Exam System (v7.3)

Field	Value
Document Version	7.3 (Enhanced Registration, Profile Management & Mobile Validation)
Date	May 15, 2026
Status	Approved for Production Deployment
Prepared For	Development Team, QA Team, UI/UX Designers, DevOps Engineers, Stakeholders
Prepared By	System Architecture & Quality Assurance Team

Table of Contents

- 1. [Introduction](#)
- 2. [Overall Description](#)
- 3. [Marketing & Feature Presentation Strategy](#)
- 4. [Landing Page & Onboarding Experience](#)
- 5. [UI/UX Design System & Color Theory](#)
- 6. [System Features \(Functional Requirements\)](#)
- 7. [External Interface Requirements](#)
- 8. [Non-Functional Requirements \(Enhanced\)](#)
- 9. [Data/Database Design](#)
- 0. [Use Cases & System Models](#)

1. [Error Handling & Edge Cases](#)
 2. [Assumptions & Dependencies](#)
 3. [Appendices](#)
-

1. Introduction

1.1 Purpose

This document provides a comprehensive specification for the **EdTech Study Platform**, a Progressive Web Application (PWA) designed for competitive exam preparation (JEE, NEET, GATE, etc.). This version (7.3) enhances the user registration and profile management systems to include mandatory Indian mobile number validation, email verification workflows, advanced password security features, and profile photo management using **Backblaze B2**. The system continues to utilize **externally managed Supabase** for backend services and **Backblaze B2** for object storage, utilizing **own credentials** for both services. This migration ensures vendor independence, eliminates AI-platform built-in dependencies, and maintains the strict "Smart Gating" and scalability requirements defined in previous iterations.

The primary purposes of this SRS are to:

- Define all functional and non-functional requirements with measurable, testable metrics.
- Establish strict acceptance criteria for quality assurance testing.
- Provide stakeholders with a clear understanding of system capabilities and constraints.
- Define the visual identity, user onboarding flow, and exam simulation environment.
- Serve as the definitive reference document for development, testing, and deployment.

Intended Audience:

- **Developers:** Implementation guidance for all system components, specifically Supabase Edge Functions, B2 integration, and enhanced auth flows.

- **QA/Test Engineers:** Test case derivation, load testing scenarios, and validation criteria.
- **UI/UX Designers:** Visual system, responsive behavior, and interaction guidelines.
- **Product Managers:** Scope definition, milestone planning, and risk assessment.
- **DevOps Engineers:** Infrastructure provisioning, monitoring, and scaling strategies.
- **Security Auditors:** Compliance verification and vulnerability assessment.

1.2 Document Conventions

Convention	Description
FR-XXX	Functional Requirement identifier
NFR-XXX	Non-Functional Requirement identifier
AC-XXX	Acceptance Criteria identifier
MR-XXX	Marketing Requirement identifier
UI-XXX	User Interface/Design Requirement identifier
EX-XXX	Enhanced Exam System Requirement identifier
Bold Text	Emphasized terms, system names, or critical constraints
Monospace	Code elements, table names, field names, API endpoints
	Locked UI state indicator
	Unlocked UI state indicator
✓	Completed state indicator

Requirement Priority Levels:

- **HIGH:** Critical to core functionality; must be implemented.
- **MEDIUM:** Important for user experience; should be implemented.
- **LOW:** Nice-to-have; can be deferred to future releases.

1.3 Intended Audience and Reading Suggestions

Reader Role	Recommended Sections
Frontend Developers	Sections 3, 4, 5, 6, 7
Backend Developers	Sections 6, 7, 8, 9
Database Engineers	Section 9
QA Engineers	Sections 6, 8, 10, 11, 13.2
UI/UX Designers	Sections 3, 4, 5, 6.5
DevOps Engineers	Sections 2.4, 2.5, 8, 13.3
Product Managers	Sections 1, 2, 3, 5, 8
Security Auditors	Sections 7.3, 8.3, 11

1.4 Project Scope

Product Vision: Build a Progressive Web App (PWA) for competitive exam preparation featuring a strict hierarchical learning path (Subject → Chapter → Topic), smart gating logic, an advanced exam simulation engine, and multimedia solution delivery (QR codes + text), all underpinned by enterprise-grade non-functional standards using **externally managed Supabase and Backblaze B2**.

Key Objectives:

- Enforce mastery-based progression through intelligent prerequisite validation.
- Integrate Telegram-based lecture delivery with internal completion tracking.
- Implement gated PDF upload system for student notes using **Backblaze B2**.
- Simulate real exam conditions with pause/resume, marking, and auto-submit.
- Provide comprehensive post-test analysis with QR-linked video solutions.
- Support 20 concurrent users initially, scalable to 2,000+ users with 99.9% availability.
- Maintain **\$0/month infrastructure cost** via Backblaze B2 Free Tier (10GB storage) + Cloudflare CDN + Supabase Free Tier.
- Maximize user acquisition via a high-conversion, responsive landing page.

- Minimize cognitive load via a specialized dark-mode-first color system.
- Ensure 99.99% data durability and <100ms API latency for core operations.
- **Strict Constraint:** No AI-platform built-in backends or storage; all services must use **own credentials**.
- **Enhanced Registration:** Mandatory Indian mobile validation (+91), email verification, and strong password policies.
- **Profile Management:** Secure photo upload/change/remove via B2, and password change functionality.

Out of Scope:

- Native mobile applications (PWA only).
- Video hosting on platform (YouTube/Telegram integration only).
- Real-time collaborative editing.
- Payment processing (initial phase).
- AI-powered content generation.

1.5 References

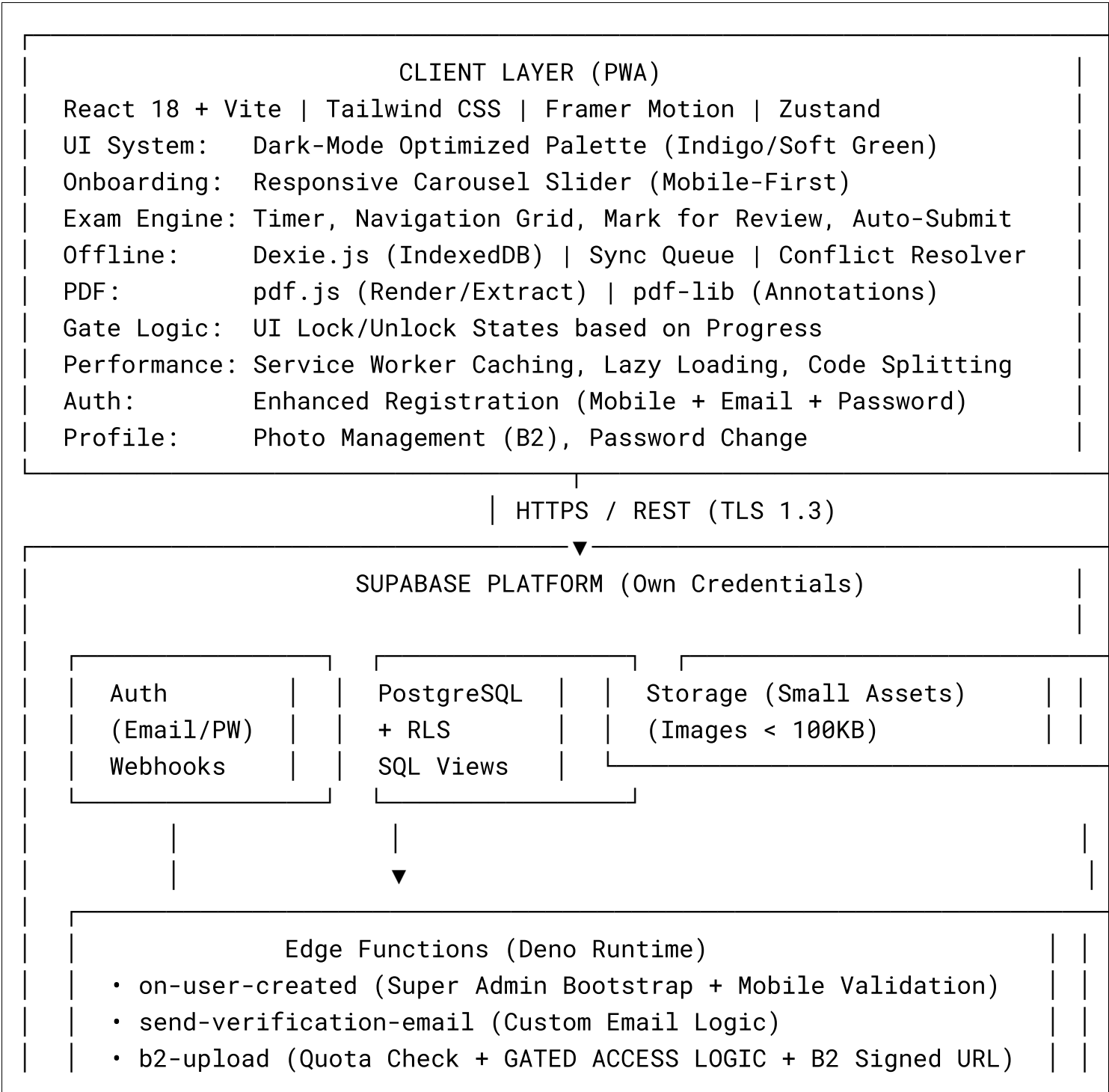
Document	URL
Supabase Documentation	Supabase Docs
Backblaze B2 Documentation	Backblaze B2 Cloud Storage Docs
Cloudflare CDN Documentation	Cloudflare Docs
IEEE Std 830-1998 (SRS Standard)	https://standards.ieee.org/
PWA Specifications	https://web.dev/progressive-web-apps/
PostgreSQL Documentation	https://www.postgresql.org/docs/
WCAG 2.1 Guidelines	https://www.w3.org/WAI/WCAG21/quickref/
OWASP Top 10	https://owasp.org/www-project-top-ten/

2. Overall Description

2.1 Product Perspective

The EdTech Study Platform is a standalone Progressive Web Application that integrates with external services for specific functionalities. The architecture emphasizes a seamless blend of rigorous backend logic, a soothing frontend, a robust exam engine, and strict adherence to non-functional quality attributes, utilizing **externally managed Supabase** and **Backblaze B2**.

System Architecture Diagram:



- b2-profile-photo (Image Compression + Upload)
- b2-download (Presigned URL for PDFs & Photos)
- storage-monitor (Daily 10GB Threshold Alert)
- gate-checker (Real-time prerequisite validation)
- exam-engine (Timer sync, Auto-submit, State management)
- qr-generator (Dynamic QR mapping for solutions)
- telegram-track (Log click events, NOT completion)
- study-planner (Auto-generate daily tasks)
- pomodoro-sync (Session logging)
- rate-limiter (Abuse prevention)

EXTERNAL SERVICES

Backblaze B2 (PDF & Profile Photo Storage - 10GB Free)
 Cloudflare CDN (Caching Layer for B2 to reduce egress)
 Telegram (Video Hosting Fallback - Deep Links Only)
 YouTube (Solution Videos - QR Linked)
 Resend (Transactional Email - Verification & Alerts)
 Sentry (Error Monitoring)

2.2 Product Functions

Core Functional Capabilities:

1. **Hierarchical Content Management:** Subject → Chapter → Topic structure with strict dependency enforcement.
2. **Smart Gating Engine:** Centralized logic layer validating prerequisite completion before unlocking content.
3. **Telegram Lecture Delivery:** Deep link integration for video content with click tracking.
4. **Gated PDF Upload System:** Student notes upload restricted to completed chapters, stored in **Backblaze B2**.
5. **Advanced Exam System:** Full simulation with timer, navigation, marking, auto-submit, and detailed analysis.
6. **Multimedia Solutions:** QR codes linking to YouTube video solutions + text explanations.

7. **Progress Tracking & Analytics:** Comprehensive monitoring of user learning journey.
8. **Pomodoro Timer Module:** Built-in focus tool with session logging and streak tracking.
9. **Smart Study Planner:** Auto-generated and manual task management system.
0. **Test Tracking:** Internal and external mock test performance logging.
1. **User Role Management:** Super Admin, Admin, Student with differentiated permissions.
2. **Offline-First Architecture:** All core features work offline with background sync.
3. **Marketing Onboarding:** Responsive slider explaining value proposition before login.
4. **Eye-Care UI:** Specialized color palette reducing strain during long study sessions.
5. **Enhanced Registration:** Mandatory Indian mobile (+91), email verification, strong password validation.
6. **User Profile Management:** Photo upload/change/remove, password change, info display.

2.3 User Classes and Characteristics

User Class	Description	Technical Proficiency	Frequency of Use
Super Admin	Full system control; manages hierarchy, configures gates, monitors storage	High (Technical)	Daily
Admin	Content creator; manages Subjects, Chapters, Topics, Tests, Solutions, QR codes	Medium-High	Daily
Student	Learner; follows gated path, takes exams, views solutions, uploads notes	Low-Medium	Multiple times daily

Student Characteristics:

- Age range: 16-25 years.
- Primary devices: Mobile (60%), Desktop (40%).

- Network conditions: Variable (3G to broadband).
- Technical literacy: Basic to intermediate.
- Motivation level: High (competitive exam preparation).
- Study Duration: Often 4-8 hours/day (requires eye-care UI).
- Exam Behavior: Needs realistic simulation (timer, navigation) to build stamina.
- **Registration:** Must provide valid Indian mobile number (+91) and verified email.

2.4 Operating Environment

Client-Side:

- **Browser Support:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+.
- **Mobile:** iOS 14+, Android 10+ (via PWA).
- **Screen Sizes:** Responsive design (320px to 2560px width).
- **Offline Capability:** IndexedDB (Dexie.js) with service worker caching.
- **Theme Support:** System-preference aware (Dark/Light), with Dark as default.
- **Hardware:** Minimum 2GB RAM, 1GHz processor.

Server-Side:

- **Database:** Supabase PostgreSQL (Free Tier initially, Own Credentials).
- **Authentication:** Supabase Auth (Email/Password, Own Credentials) with custom mobile validation.
- **Edge Functions:** Deno Runtime on Supabase Edge (Own Credentials).
- **Storage:** Backblaze B2 (10GB free tier, target 9GB usage, Own Credentials) for PDFs and Profile Photos.
- **CDN:** Cloudflare CDN in front of B2 to cache content and reduce egress.
- **Hosting:** Vercel for frontend deployment.
- **Monitoring:** Sentry for error tracking, Supabase Dashboard for logs.

Third-Party Services:

- Telegram (lecture delivery via deep links).
- YouTube (solution videos).
- Resend (transactional emails, including verification).
- Cloudflare (CDN).

2.5 Design and Implementation Constraints

ID	Description	Rationale
C-01	\$0/month infrastructure cost for initial 20 users	Budget limitation for MVP
C-02	Maximum 9GB Backblaze B2 usage	Free tier limitation (10GB total)
C-03	500MB per-user PDF storage limit	Prevent abuse and control costs
C-04	10MB maximum file size per PDF upload	Bandwidth and processing constraints
C-05	No native mobile apps (PWA only)	Development timeline and cost
C-06	Telegram used for delivery only, not completion tracking	API limitations and reliability
C-07	Offline-first architecture required	User accessibility in low-connectivity areas
C-08	Supabase Free Tier row limits (500K rows)	Database scaling constraint
C-09	Gate check latency < 50ms	User experience requirement
C-10	Page load time < 2s on 3G networks	Accessibility requirement
C-11	UI must use Dark Mode as primary theme	Reduce eye strain for long study sessions
C-12	Landing page must be fully responsive (Mobile First)	Primary traffic source is mobile
C-13	Exam timer must sync server-client to prevent cheating	Integrity of assessment
C-14	QR codes must map to specific Question/Topic levels	Granular solution delivery

ID	Description	Rationale
C-15	Zero data loss during exam submissions	Critical reliability requirement
C-16	99.9% uptime during exam windows	Availability requirement
C-17	No AI-platform built-in backends	Mandatory external infrastructure
C-18	Own Credentials Required	Supabase and B2 must use user-provided keys
C-19	Indian Mobile Validation	Mandatory +91 prefix, 10 digits, regex validation
C-20	Email Verification Mandatory	Login blocked until email verified
C-21	Profile Photo Storage	Must use Backblaze B2 with compression

2.6 Assumptions and Dependencies

Assumptions:

- Users have reliable internet access for initial content download and sync operations.
- Telegram deep links function correctly on all target platforms.
- Supabase Free Tier remains available with current specifications.
- Backblaze B2 free tier (10GB) remains unchanged.
- Students possess basic device literacy for PWA installation.
- Email delivery via Resend achieves >95% deliverability.
- Users will not attempt systematic gate bypass attacks.
- Students prefer dark interfaces for night-time study sessions.
- YouTube videos for solutions are publicly accessible or unlisted.
- Network latency between user and edge functions is < 100ms.
- Cloudflare CDN effectively caches B2 content to minimize egress fees.
- Users possess valid Indian mobile numbers (+91) for registration.

Dependencies:

- **Supabase:** Authentication, Database, Storage (small assets), Edge Functions (Own Credentials).
 - **Backblaze B2:** Object storage for PDF files and Profile Photos (Own Credentials).
 - **Cloudflare:** CDN layer for B2 to optimize egress.
 - **Telegram:** Video content hosting and deep link delivery.
 - **Vercel:** Frontend hosting and CDN.
 - **Resend:** Transactional email delivery (Verification, Welcome, Alerts).
 - **Node.js v18+:** Development environment requirement.
 - **Git:** Version control system.
 - **YouTube:** Video hosting for solutions.
 - **Sentry:** Error monitoring and crash reporting.
-

3. Marketing & Feature Presentation Strategy

3.1 High-Impact Feature Extraction

To drive user conversion, the following 5 features are highlighted on the landing page and marketing materials. These are selected for their immediate value to students and visual representability.

Feature Title	One-Line Description	Tagline	Visual Idea
Smart Mastery Path	"No shortcuts: Unlock new topics only after mastering the previous ones."	<i>Learn Deep, Not Fast.</i>	A locked padlock turning into a green checkmark on a progress tree.
Real Exam Simulation	"Practice with full-screen timers, navigation grids, and auto-submit just like the real test."	<i>Train Like You Fight.</i>	A split screen showing a timer and a question grid.
Instant Video Solutions	"Scan a QR code to watch expert video explanations for every question."	<i>See It, Solve It.</i>	A smartphone scanning a QR code that reveals a video player.

Feature Title	One-Line Description	Tagline	Visual Idea
Distraction-Free Focus	"Built-in Pomodoro timer and task planner to keep you on track."	Stay in the Zone.	A floating timer widget overlaying a clean study interface.
Smart Note Vault	"Upload your notes only after completing a chapter to reinforce learning."	Earn Your Notes.	A folder icon unlocking with a glowing effect after a test is passed.

3.2 Marketing Messaging Guidelines

- **Tone:** Encouraging, disciplined, professional, and empathetic to student stress.
- **Keywords:** Mastery, Focus, Smart Gating, Analytics, Distraction-Free, Real Exam, Video Solutions, Verified Account.
- **Avoid:** "Easy pass," "Cheat codes," or anything suggesting the system can be bypassed.

4. Landing Page & Onboarding Experience

4.1 Responsive Landing Page Slider Design

A modern, minimal, distraction-free carousel slider serves as the primary onboarding experience. It introduces the platform's value proposition before the user signs up.

Structure:

Slide	Content
Screen 1	Smart Mastery Path (Feature 1)
Screen 2	Real Exam Simulation (Feature 2)

Slide	Content
Screen 3	Instant Video Solutions (Feature 3)
Screen 4	Distraction-Free Focus (Feature 4)
Screen 5 (Final)	Strong CTA + Login/Sign Up

4.2 Responsive Behavior Specifications

4.2.1 Layout System

- **Grid/Flex:** CSS Flexbox for vertical stacking on mobile, horizontal alignment on desktop.
- **Breakpoints:**
 - Mobile: < 768px (Vertical Stack)
 - Tablet: 768px - 1024px (Balanced)
 - Desktop: > 1024px (Horizontal Split)

4.2.2 Screen-by-Screen Breakdown

Screen	Mobile	Tablet	Desktop
1: Smart Mastery Path	Icon top, Title, Description, "Next" arrow bottom	Icon left (30%), Content right (70%)	Icon left (20%), Content center (60%), Empty space right (20%)
2: Real Exam Simulation	Stacked vertically. Visual of timer and question grid	Graphic left, text right	Graphic floating left, text centered
3: Instant Video Solutions	Stacked. Visual of phone scanning QR code	Visual of QR expanding into video player	Visual of QR expanding into video player
4: Distraction-Free Focus	Stacked. Timer graphic prominent	Graphic left, text right	Graphic floating left, text centered

Screen	Mobile	Tablet	Desktop
5: Call to Action (CTA)	"Get Started" full width. "Login" link below	"Get Started" centered. "Already have an account? Login" below	"Get Started" prominent. "Login" link subtle

4.3 UX & Interaction Details

Button Microcopy:

- **Primary:** "Get Started", "Continue", "Start Studying", "Start Exam".
- **Secondary:** "Skip", "Later", "Login", "Pause Exam".
- **Loading:** "Creating Account...", "Loading...", "Starting Timer...", "Verifying Email...".

Progress Indicators:

- **Dots:** 5 dots at the bottom. Active dot is filled (Primary Color), inactive is outline (Secondary Color).
- **Swipe:** Horizontal swipe gesture enabled on mobile.
- **Transitions:** Smooth fade-in/slide-up animation (300ms ease-out) between slides.

CTA Placement Strategy:

- **Mobile:** Fixed at bottom of viewport (sticky) for easy thumb reach.
- **Desktop:** Centered in the visual focal point.

5. UI/UX Design System & Color Theory

5.1 Design Philosophy

The UI is optimized for long-duration study sessions (4-8 hours) and high-focus exam simulations. The primary goal is to reduce eye strain, maintain focus, and provide clear visual hierarchy without overwhelming the user.

Core Principles:

- **Dark Mode First:** Default theme is dark to reduce blue light emission and glare.
- **Low Contrast Text:** Avoid pure white text on pure black backgrounds.
- **Calm Accents:** Use cool tones (Indigo, Blue, Soft Green) to promote focus.
- **Minimalist:** Reduce visual noise; hide non-essential elements during exams.
- **Accessibility:** WCAG 2.1 AA compliant contrast ratios.

5.2 Color Palette (Exact HEX Codes)

Role	Color Name	Hex Code	Usage
Background	Deep Slate	#0F172A	Main app background (Dark Mode)
Surface/Card	Midnight Blue	#1E293B	Cards, modals, input fields
Surface/Light	Soft Gray	#F1F5F9	Background (Light Mode)
Primary	Focus Indigo	#6366F1	Primary buttons, active states, links
Primary Hover	Deep Indigo	#4F46E5	Hover states for primary elements
Secondary	Calm Teal	#14B8A6	Secondary actions, success indicators
Accent	Soft Purple	#A855F7	Highlights, badges, special tags
Text Primary	Off-White	#F8FAFC	Headings, main body text (Dark Mode)
Text Secondary	Slate Gray	#94A3B8	Subtitles, labels, disabled text
Text Disabled	Dim Gray	#475569	Inactive elements
Border/Divider	Muted Border	#334155	Separators, input borders
Success	Growth Green	#10B981	Completed tasks, passed quizzes, correct answers

Role	Color Name	Hex Code	Usage
Warning	Caution Amber	#F59E0B	Warnings, approaching limits, marked for review
Error	Alert Red	#EF4444	Errors, failed attempts, bans, incorrect answers

5.3 Color Usage Guidelines

Accessibility & Contrast:

- **Text on Background:** Ensure contrast ratio > 4.5:1 (WCAG AA).
- **Text on Surface:** Ensure contrast ratio > 3:1 for large text.
- **Avoid:** Pure Black (#000000) and Pure White (#FFFFFF) to prevent "halation" effect.

Do's:

- Use Focus Indigo for primary actions only.
- Use Off-White for reading text.
- Use Midnight Blue for cards to create depth.
- Use Growth Green sparingly to celebrate achievements.
- **Exam Mode:** Use high-contrast Alert Red and Growth Green for answer status.

Don'ts:

- Do not use bright red for non-critical warnings.
- Do not use pure white backgrounds in Dark Mode.
- Do not use more than 2 accent colors on a single screen.

5.4 Component-Level Styling

Component	Color Usage
Primary Button	BG: Focus Indigo, Text: Off-White, Hover: Deep Indigo
Secondary Button	BG: Transparent, Border: Muted Border, Text: Slate Gray

Component	Color Usage
Disabled Button	BG: Dim Gray, Text: Dim Gray, Cursor: Not Allowed
Navigation Bar	BG: Deep Slate, Border Bottom: Muted Border
Subject Card	BG: Midnight Blue, Border: Muted Border, Icon: Focus Indigo
Locked State	Opacity: 0.6, Filter: Grayscale, Icon: Dim Gray
Unlocked State	Opacity: 1.0, Border: Focus Indigo (subtle glow)
Progress Bar	BG: Muted Border, Fill: Focus Indigo
Alert Banner	BG: Alert Red (low opacity), Text: Alert Red
Success Banner	BG: Growth Green (low opacity), Text: Growth Green
Exam Timer	BG: Midnight Blue, Text: Off-White, Warning: Caution Amber
Question Grid	Active: Focus Indigo, Correct: Growth Green, Incorrect: Alert Red, Marked: Caution Amber
Profile Avatar	Circular, Border: Focus Indigo (if verified), Gray (if pending)

5.5 Theme Strategy

- **Default:** Dark Mode (Deep Slate background).
- **Light Mode:** Available via settings (Soft Gray background).
- **Adaptability:** All colors use CSS variables (e.g., `--bg-primary`, `--text-main`) to ensure seamless switching.
- **Cross-Device:** Colors render consistently across iOS, Android, and Desktop browsers.

6. System Features (Functional Requirements)

6.1 Hierarchical Content Structure

FR-H01: Strict Hierarchy Enforcement

Priority: HIGH | **Status:** Approved

Description: System must enforce a rigid Subject → Chapter → Topic dependency chain.

Requirements:

- A Topic cannot exist without a parent Chapter.
- A Chapter cannot exist without a parent Subject.
- Content per Topic includes:
 - Lecture (Telegram deep link)
 - Lecture Quiz (mandatory pass to proceed)
 - DPP (Daily Practice Problems) – unlocks after Lecture Quiz pass
 - PYQs (Previous Year Questions) – unlocks after DPP completion
 - Topic Test – unlocks after PYQs completion (marks Topic Complete)

Acceptance Criteria:

ID	Criterion
AC-H01	Attempting to create a Topic without a Chapter results in database error.
AC-H02	Attempting to create a Chapter without a Subject results in database error.

FR-H02: Chapter & Subject Progression

Priority: HIGH | **Status:** Approved

Description: System must track and enforce completion requirements at chapter and subject levels.

Requirements:

- **Chapter Completion:** Requires ALL Topics in the chapter to be marked "Complete". Unlocks: Chapter Test and Chapter PYQs Test.
- **Subject Completion:** Requires ALL Chapters in the subject to be marked "Complete". Unlocks: Subject Test and Subject PYQs Test.

Acceptance Criteria:

ID	Criterion
AC-H03	Chapter Test remains locked until all topics in chapter are complete.
AC-H04	Subject Test remains locked until all chapters in subject are complete.

FR-H03: Multi-Level Test Unlocking

Priority: MEDIUM | **Status:** Approved

Description: System must unlock advanced tests based on cumulative subject completion.

Requirements:

- **Multi-Subject Tests:** Unlocks when ≥ 2 Subjects are completed.
- **Full Syllabus / Grand Tests:** Unlocks when ALL Subjects are completed.

Acceptance Criteria:

ID	Criterion
AC-H05	Multi-Subject Test accessible with 2+ subjects complete.
AC-H06	Grand Test accessible only when all subjects complete.

6.2 Telegram-Based Lecture Delivery

FR-T01: Delivery Mechanism

Priority: HIGH | **Status:** Approved

Description: Telegram used solely for content distribution via deep links.

Requirements:

- Admins store telegram_chat_id and telegram_message_id for each Topic.
- Student clicks "Watch Lecture" → Opens Telegram app via deep link (tg://resolve?domain=...).
- System logs lecture_click_event (timestamp, user_id) in database.
- **Constraint:** NO reliance on Telegram API for "completion" status.
- Completion determined solely by passing the internal Lecture Quiz.

Acceptance Criteria:

ID	Criterion
AC-T01	Clicking "Watch Lecture" opens Telegram app to correct message.
AC-T02	Lecture click logged in lecture_clicks table.
AC-T03	No automatic "Completion" status set by Telegram click.
AC-T04	Completion status set only after passing internal Lecture Quiz.

FR-T02: Deep Linking & Analytics

Priority: MEDIUM | **Status:** Approved

Description: System must track lecture engagement metrics.

Requirements:

- Track click-through rates per lecture.
- Flag lectures with <10% click-through for admin review.
- Display analytics in admin dashboard.

Acceptance Criteria:

ID	Criterion
AC-T05	Admin dashboard displays lecture click-through rates.
AC-T06	Lectures with <10% CTR flagged for review.

6.3 Smart Gating System (Core Logic)

FR-G01: Prerequisite Validation Engine

Priority: HIGH | Status: Approved

Description: System must enforce "No Quiz Pass → No DPP → No PYQ → No Test → No Chapter Completion → No PDF Upload" rule.

Requirements:

- **Frontend:** UI components check gate_status before rendering interactive elements.
- **Backend:** API endpoints reject requests if prerequisites not met (HTTP 403).
- **Database:** Triggers validate prerequisite_passed flag before allowing state changes.

Acceptance Criteria:

ID	Criterion
AC-G01	DPP inaccessible if Lecture Quiz not passed (API returns 403).
AC-G02	PYQs inaccessible if DPP not completed (API returns 403).
AC-G03	Topic Test inaccessible if PYQs not completed (API returns 403).
AC-G04	PDF Upload inaccessible if Chapter Test not attempted (API returns 403).

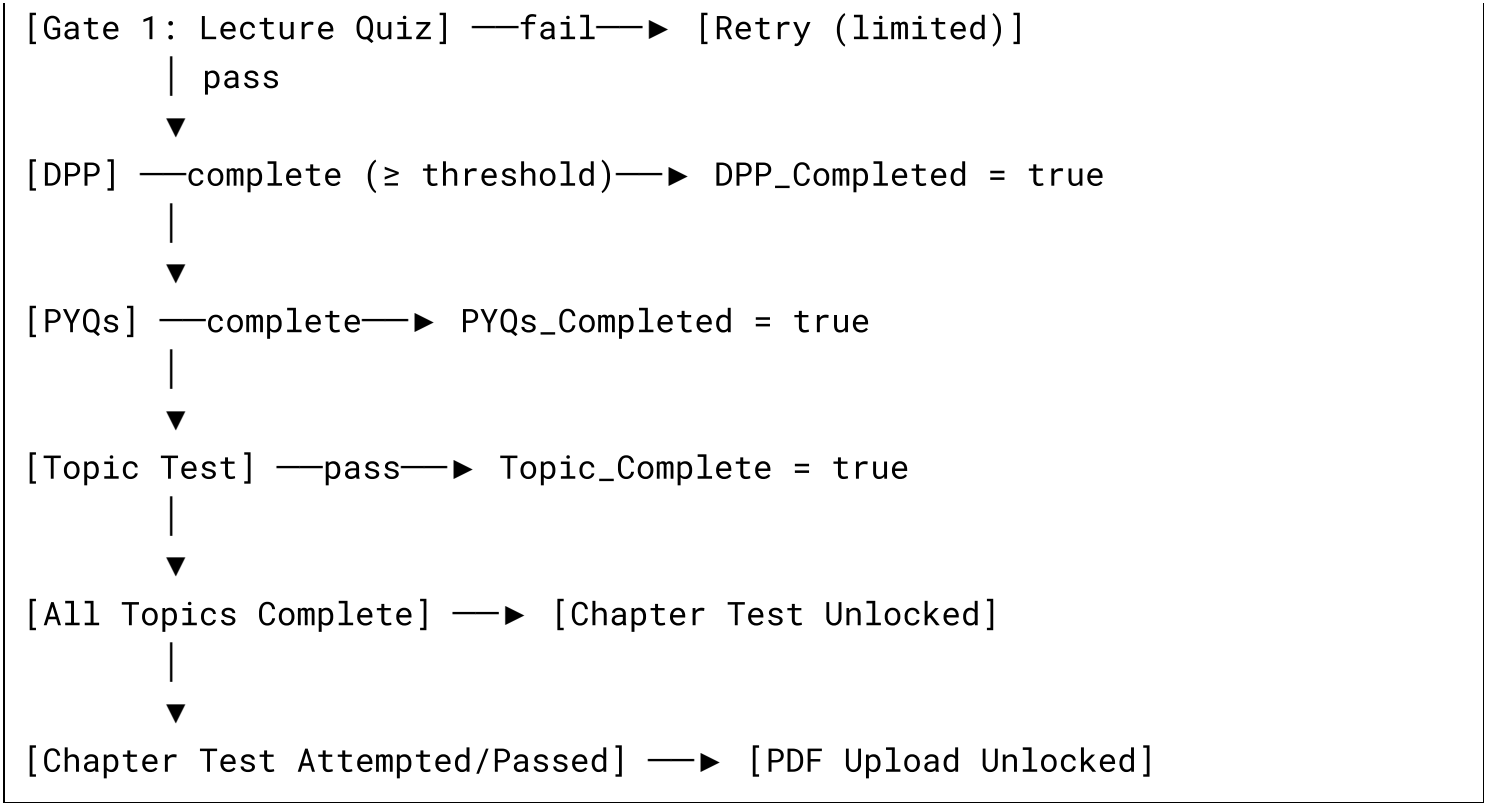
FR-G02: Gating Flow Logic

Priority: HIGH | Status: Approved

Description: Detailed step-by-step gating workflow.

Gating Flow Diagram:





Acceptance Criteria:

ID	Criterion
AC-G05	Each gate transition properly recorded in progress tables.
AC-G06	Configurable retry limits enforced for quizzes.

6.4 Student PDF Upload System (Gated Feature - Backblaze B2)

FR-P01: Gated Upload Access

Priority: HIGH | **Status:** Approved

Description: Student can upload PDFs ONLY after chapter completion. Storage is handled by **Backblaze B2**.

Requirements:

- **Condition:** chapter_completion_status == 'completed' (All topics done + Chapter Test attempted).
- **Scope:** Uploads scoped to specific Chapter or Topic being studied.
- **UI State:**
 -  **Locked:** "Complete Chapter to unlock Notes Upload" (Motivational trigger).
 -  **Unlocked:** "Upload Your Notes" button active.
- **Storage Provider:** Backblaze B2 (Own Credentials).

Acceptance Criteria:

ID	Criterion
AC-P01	User sees "Locked" state for PDF upload before Chapter completion.
AC-P02	User sees "Unlocked" state for PDF upload after Chapter Test.
AC-P03	Upload fails if global B2 usage > 9GB (returns 429).
AC-P04	Upload fails if user exceeds 500MB personal limit.

FR-P02: Storage & Optimization (Backblaze B2)

Priority: HIGH | **Status:** Approved

Description: PDF storage must be optimized, searchable, and secured via **Backblaze B2**.

Requirements:

Setting	Value
Global Limit	≤ 9GB total B2 usage
Per-User Limit	500MB (Hard Limit)
File Size	Max 10MB per file
Processing	Client-side pdf.js extracts text for search
Annotations	Highlights/Notes stored as JSON
Immutability	PDF content cannot be edited; re-upload required
Storage Path	users/{uid}/chapters/{chapter_id}/notes_{timestamp}.pdf
CDN	Cloudflare CDN in front of B2 for caching

Acceptance Criteria:

ID	Criterion
AC-P05	PDF text extracted and searchable.
AC-P06	Deleting a note removes PDF from B2.
AC-P07	Annotations persist across sessions.
AC-P08	B2 Signed URLs expire after 15 minutes for security.
AC-P09	CDN caching reduces B2 egress by >80%.

6.5 Advanced Test & Exam System (Enhanced)

EX-01: Test Structure & Hierarchy

Priority: HIGH | Status: Approved

Description: Tests must follow the Subject → Chapter → Topic hierarchy with specific types.

Test Types:

Type	Scope
Topic-wise Tests	Focused on a single topic
Chapter-wise Tests	Cumulative test for a chapter
Subject-wise Tests	Cumulative test for a subject
DPP (Daily Practice Problems)	Short, daily problem sets

- **Mapping:** Each test is linked to a specific hierarchy level (Topic, Chapter, or Subject).
- **Unlocking:** Tests unlock based on the Smart Gating System (Section 6.3).

Acceptance Criteria:

ID	Criterion
AC-EX01	Tests are correctly categorized by type and hierarchy level.
AC-EX02	DPPs are distinct from formal Tests but follow similar gating logic.
AC-EX03	Admins can assign tests to specific hierarchy nodes.

EX-02: Exam Attempt Functionality

Priority: HIGH | Status: Approved

Description: The exam interface must simulate a real testing environment with robust controls.

Requirements:

- **Start / Pause / Resume:**
 - Users can start an exam.
 - **Pause:** Limited pauses allowed (e.g., 2 per exam). Timer stops.
 - **Resume:** Users can resume from the exact state.
 - **Auto-Submit:** If timer expires or user closes the tab without pausing (configurable), exam auto-submits with current answers.
- **Question Navigation:**

- **Grid View:** Sidebar/grid showing all question numbers.
- **Status Indicators:**

Color	Status
Green	Answered
Red	Not Answered
Yellow	Marked for Review
Gray	Not Visited

- **Direct Jump:** Clicking a number jumps to that question immediately.
- **Mark for Review:** Users can flag questions to revisit later. Flagged questions are highlighted in the grid.
- **Timer System:**
 - **Countdown:** Visible timer (HH:MM:SS).
 - **Warnings:** Visual/audible warnings at 5 mins and 1 min remaining.
 - **Sync:** Server-side time sync to prevent client-side manipulation.
- **Auto-Submit Logic:**
 - Triggered on timeout.
 - Triggered if user attempts to navigate away (with a "Save & Exit" confirmation).
 - Final submission locks the attempt; no further changes allowed.

Acceptance Criteria:

ID	Criterion
AC-EX04	Timer pauses correctly and resumes accurately.
AC-EX05	Auto-submit triggers exactly at 00:00.
AC-EX06	Question grid updates status in real-time as user interacts.
AC-EX07	"Mark for Review" toggles state and updates grid color.

EX-03: Result & Analysis Engine

Priority: HIGH | **Status:** Approved

Description: Immediate and detailed feedback post-exam.

Requirements:

- **Score Calculation:**
 - **Marks:** Total marks obtained.
 - **Accuracy:** Percentage of correct answers.
 - **Percentile:** (Optional) Calculated against other users who took the same test.
 - **Negative Marking:** Applied if configured (e.g., -1 for wrong answer).
- **Question-Wise Status:** Correct, Incorrect, Skipped, Not Attempted; user's selected option vs. correct option.
- **Time Analytics:**
 - **Time Spent:** Duration spent on each question.
 - **Average Time:** Comparison with class average (if data available).
- **Performance Insights:**
 - **Strengths/Weaknesses:** Identification of topics with low accuracy.
 - **Speed Analysis:** "You spent 2x longer on Physics questions."
 - **Recommendations:** "Revise Topic X based on errors."

Acceptance Criteria:

ID	Criterion
AC-EX08	Score is calculated instantly upon submission.
AC-EX09	Negative marking is applied correctly if enabled.
AC-EX10	Time-per-question data is logged and displayed.
AC-EX11	Insights are generated based on configurable logic.

EX-04: Answer Sheet System

Priority: HIGH | **Status:** Approved

Description: A comprehensive view of the user's attempt with solutions.

Requirements:

- **Answer Sheet View:** Displays the full question paper with user's answers highlighted; shows the correct answer key side-by-side.
- **Availability:** ONLY for DPPs and Tests (not for simple quizzes).
- **Detailed Solutions:**

- **Text Explanation:** Step-by-step text solution for each question.
- **Formatting:** Supports LaTeX/MathJax for equations.
- **Availability:** Only unlocked after exam submission.
- **Navigation:** Scrollable list or paginated view; filter by "Incorrect Only" or "Skipped Only".

Acceptance Criteria:

ID	Criterion
AC-EX12	Answer sheet is inaccessible before exam submission.
AC-EX13	User answers are clearly distinguished from correct answers.
AC-EX14	Text solutions render correctly with mathematical symbols.

EX-05: QR Code-Based Video Solutions

Priority: MEDIUM | **Status:** Approved

Description: Integration of video solutions via QR codes for offline/quick access.

Requirements:

- **QR Code Generation – Levels:**

Level	Purpose
Subject	Overview video
Chapter	Chapter summary video
Topic	Topic-specific concept video
Question	Specific solution video for a question

- **Mapping:** Admins attach YouTube video URLs to specific hierarchy levels or individual questions; system encodes these URLs into QR codes.
- **User Flow:** User scans QR (via camera or in-app scanner) → Redirects to YouTube video.
- **Fallback:** If QR scan fails or network is poor, a clickable link/button is shown.

- **Display:** QR codes displayed on the Answer Sheet next to the question; also available in the "Solutions" tab.

Acceptance Criteria:

ID	Criterion
AC-EX15	QR codes generate valid URLs pointing to correct videos.
AC-EX16	Scanning a QR code opens the correct video immediately.
AC-EX17	Fallback text link is visible if video is unavailable.

EX-06: Content Mapping Rules

Priority: HIGH | **Status:** Approved

Description: Strict rules for linking content, solutions, and media.

Requirements:

- **Hierarchy Linking:**
 - Question ↔ Solution (Text) ↔ Solution (Video/QR).
 - Topic ↔ Topic Video ↔ Topic Test.
- **Access Control:**
 - Answer Sheets and Solutions are only available for DPPs and Tests.
 - Quizzes (Lecture Quiz) may have immediate feedback but not full answer sheets.
- **Consistency:** If a video is missing for a question, system must show "Video Solution Unavailable" but still show the text solution.

Acceptance Criteria:

ID	Criterion
AC-EX18	No answer sheets are generated for non-test content.
AC-EX19	Missing videos do not break the UI; fallback is graceful.

EX-07: Admin/Content Management for Exams

Priority: HIGH | **Status:** Approved

Description: Tools for admins to create and manage the exam ecosystem.

Requirements:

- **Test Creation:** Create Tests/DPPs by selecting questions from a pool; set duration, negative marking, and passing score.
- **Solution Upload:** Upload text solutions (Markdown/LaTeX); paste YouTube URLs for video solutions; system auto-generates QR codes.
- **Answer Sheet Management:** Review and publish answer sheets (optional moderation); bulk upload answer keys via CSV.
- **Mapping Interface:** Drag-and-drop interface to link videos to specific questions/topics.

Acceptance Criteria:

ID	Criterion
AC-EX20	Admins can create a test with 100+ questions efficiently.
AC-EX21	QR codes are generated automatically upon saving a video URL.
AC-EX22	Bulk upload validates question IDs and hierarchy.

EX-08: UI/UX for Exam Interface

Priority: HIGH | **Status:** Approved

Description: A clean, distraction-free interface optimized for focus.

Requirements:

- **Layout:**
 - Left/Top: Question text and options.
 - Right/Bottom: Question Grid (Navigation).
 - Header: Timer, User Info, "Save & Next", "Mark for Review", "Submit".
- **Responsiveness:**
 - **Mobile:** Stacked layout (Question top, Grid bottom). Swipe to navigate.
 - **Desktop:** Split screen (Question left, Grid right).
- **Interactions:** Smooth transitions; no page reloads (SPA); Keyboard Shortcuts (Desktop): Arrow keys to navigate, M to mark, Enter to save.

- **Result Dashboard:** Visual charts (Pie chart for accuracy, Bar chart for topic-wise performance); Tabs for "Score", "Analysis", "Answer Sheet", "Video Solutions".

Acceptance Criteria:

ID	Criterion
AC-EX23	Exam interface loads in < 1s.
AC-EX24	Mobile layout is touch-friendly (large tap targets).
AC-EX25	Keyboard shortcuts work without interfering with typing.

EX-09: Edge Cases & Error Handling

Priority: HIGH | **Status:** Approved

Description: Robust handling of unexpected scenarios during exams.

Requirements:

- **Missing QR/Video:** Show placeholder "Video solution coming soon"; do not block access to text solution.
- **Network Issues:**
 - **Offline Mode:** Save answers locally (IndexedDB).
 - **Reconnect:** Auto-sync answers when connection is restored.
 - **Timeout:** Disconnected > 5 mins → warn user; > 10 mins → auto-submit.
- **Anti-Cheating (Basic):**
 - **Tab Switching:** Warn user (configurable).
 - **Copy/Paste:** Disable in question text area (optional).
 - **Screenshot:** (Future) Detect screenshot attempts.
- **Browser Crash:** User can resume the exam from the last saved state upon reopening.

Acceptance Criteria:

ID	Criterion
AC-EX26	Local storage saves every answer change.

ID	Criterion
AC-EX27	Auto-submit triggers gracefully on prolonged disconnection.
AC-EX28	Tab-switch warnings are non-intrusive but clear.

6.6 Tracking & Analytics

FR-A01: Comprehensive Tracking

Priority: HIGH | **Status:** Approved

Description: System must track all user interactions and performance metrics.

Tracked Item	Data Captured
Lecture	Click events (not completion)
Quizzes / DPPs / PYQs	Accuracy, Time Taken, Attempts, Score
Tests	Score, Percentile, Time, Weak Areas, Question-level time
Progress	Subject_Chapter_Topic completion tree
Exam Behavior	Pause frequency, time spent on "Marked" questions

Acceptance Criteria:

ID	Criterion
AC-A01	All interaction types logged with timestamps.
AC-A02	Performance metrics accurately calculated.

FR-A02: Dashboard Visualization

Priority: MEDIUM | **Status:** Approved

Description: User dashboard must display progress and next actions.

Requirements:

- **Current Progress:** Visual tree showing 🔒 Locked / 🔓 Unlocked / ✓ Completed.
- **Next Action:** "Complete Lecture Quiz for Topic X".
- **Weak Areas:** Heatmap of low-accuracy topics.
- **Upload Status:** "Chapter Y Complete - Notes Upload Unlocked".
- **Exam Stats:** "Last Test: 85% | Avg Time: 45s/question".

Acceptance Criteria:

ID	Criterion
AC-A03	Dashboard accurately reflects Locked/Unlocked/Completed states.
AC-A04	"Next Action" card correctly identifies next gated step.
AC-A05	Weak area heatmap updates based on quiz/DPP/test accuracy.

6.7 Admin Features

FR-AD01: Hierarchy Management

Priority: HIGH | Status: Approved

Description: Admins must be able to manage content hierarchy.

Requirements:

- CRUD for Subjects, Chapters, Topics.
- Bulk import for Questions (DPP, PYQ, Tests).
- Configure Telegram deep links for lectures.
- **Exam Specific:** Manage test pools, set negative marking rules.

Acceptance Criteria:

ID	Criterion
AC-AD01	Admin can create/edit/delete hierarchy elements.
AC-AD02	Bulk import validates hierarchy before insertion.

FR-AD02: Gate Configuration

Priority: MEDIUM | **Status:** Approved

Description: Admins must configure gating rules without code changes.

Requirements:

- Set thresholds (e.g., "Must score 70% on Lecture Quiz").
- Set retry limits.
- Configure difficulty levels.
- Changes apply immediately to all future attempts.

Acceptance Criteria:

ID	Criterion
AC-AD03	Gate configuration changes reflected immediately.
AC-AD04	Configuration persisted across system restarts.

FR-AD03: User Management & Moderation

Priority: HIGH | **Status:** Approved

Description: Admins must manage user accounts and content.

Requirements:

- **Approval Queue:** List of pending_approval users with "Approve" / "Reject" buttons.
- **Ban/Suspend:** Ability to suspend (blocks login) or ban (blocks account) a user.
- **Reset Progress:** super_admin can reset user's progress for specific Topic/Chapter/Test.
- **View User Data:** super_admin can view user's notes/PDFs/exam attempts for moderation.

Acceptance Criteria:

ID	Criterion
AC-AD05	User approval workflow functional.
AC-AD06	Suspended users cannot access system.

ID	Criterion
AC-AD07	Progress reset properly clears related data.

6.8 System Bootstrap & Role Management

FR-B00T-01: Automated Super Admin Bootstrap

Priority: HIGH | **Status:** Approved

Description: First registered user automatically becomes Super Admin.

Requirements:

- **Trigger:** Immediately upon first successful user registration and email verification.
- **Logic:**
 - System checks profiles table count.
 - **If Count == 0:** Assign role = 'super_admin'; set is_approved = true; set status = 'approved'; log "System Bootstrapped: User [ID] is Super Admin".
 - **If Count > 0:** Assign role = 'student'; set is_approved = false; set status = 'pending_approval'.
- **Implementation:** Supabase Auth Webhook (on-user-created) + Database Trigger fallback.
- **Security Constraint:** No other user can be assigned super_admin via registration.

Acceptance Criteria:

ID	Criterion
AC-B01	First user becomes Super Admin automatically.
AC-B02	Second user is Pending Approval.
AC-B03	Race condition prevented via database transaction.

FR-B00T-02: Role Promotion & Demotion Workflow

Priority: HIGH | **Status:** Approved

Description: Role changes must be controlled and audited.

Requirements:

- **Promotion:**
 - super_admin can promote any student or admin to admin or super_admin.
 - **Constraint:** A super_admin cannot demote themselves to lower role (prevent lockout).
- **Demotion:**
 - super_admin can demote admin to student.
 - super_admin can demote super_admin to admin (if multiple admins exist).
- **Audit Log:** Every role change logged in audit_logs.

Acceptance Criteria:

ID	Criterion
AC-B04	Role promotions require super_admin authorization.
AC-B05	Self-demotion blocked for sole super_admin.
AC-B06	All role changes logged with timestamp and actor.

FR-B00T-03: Emergency Access Recovery

Priority: MEDIUM | **Status:** Approved

Description: Recovery mechanism for lost super_admin access.

Requirements:

- **Option A:** Manual SQL intervention by hosting provider (identity verified).
- **Option B (Preferred):** Pre-configured "Recovery Code" via secure Edge Function.
 - Recovery code stored securely by founder.
 - Can reset super_admin role for specific user ID.

Acceptance Criteria:

ID	Criterion
AC-B07	Recovery mechanism tested and documented.
AC-B08	Recovery code protected from unauthorized access.

6.9 Pomodoro Technique Module

FR-POM0-01: Built-in Timer

Priority: MEDIUM | Status: Approved

Description: Integrated focus timer for study sessions.

Requirements:

- **Modes:** Standard (25m Work / 5m Break), Long Break (15m), Custom.
- **State Management:** Timer state persists across page refreshes (IndexedDB + Server Sync).
- **Audio:** Gentle chime for start/end/break.
- **Background:** Works in background (Service Worker) on supported browsers.

Acceptance Criteria:

ID	Criterion
AC-P01	Timer works offline and syncs on reconnect.
AC-P02	Background timer continues on supported browsers.

FR-POM0-02: Session Logging & Streaks

Priority: MEDIUM | Status: Approved

Description: Track study sessions and gamify engagement.

Requirements:

- **Logging:** Every completed session logs to pomodoro_sessions table (user_id, duration_seconds, topic_context (optional), start_time, end_time).

- **Streaks:** Calculates daily/weekly streaks based on minimum sessions (e.g., 4 sessions/day).
- **Gamification:** "Focus Master" badge for 7-day streak.

Acceptance Criteria:

ID	Criterion
AC-P03	Sessions logged accurately with timestamps.
AC-P04	"Focus Streak" increments only after completing full sessions.

FR-POM0-03: Integration with Study Plan

Priority: LOW | **Status:** Approved

Description: Connect timer activity with study planning.

Requirements:

- **Context Awareness:** Timer started while viewing Topic tags session to that Topic.
- **Dashboard:** Shows "Total Focus Time Today" and "Current Streak".

Acceptance Criteria:

ID	Criterion
AC-P05	Session context properly associated with topic.
AC-P06	Dashboard displays accurate focus metrics.

6.10 Smart Study Planner

FR-PLAN-01: Dual-Mode Task System

Priority: MEDIUM | **Status:** Approved

Description: Auto-generated and manual task management.

Requirements:

- **Auto-Generated Tasks:**

- **Trigger:** Daily cron job (study-planner Edge Function) at 00:00 UTC.
- **Logic:** Fetch v_weak_topics (Topics with <50% accuracy in last 3 attempts); fetch due_revision_tasks (Spaced Repetition due today); generate 3-5 suggested tasks for the day; insert into study_tasks with source = 'auto'.
- **Manual Tasks:**
 - User can create custom tasks (e.g., "Read Chapter 5", "Revise Formulas"). source = 'manual'.
 - Can link to platform_subtopic_id, personal_topic_id, or free text.

Acceptance Criteria:

ID	Criterion
AC-PL01	Auto-tasks generated daily at midnight UTC.
AC-PL02	Manual tasks creatable with various target types.

FR-PLAN-02: Task Lifecycle

Priority: MEDIUM | **Status:** Approved

Description: Tasks must have clear status transitions.

Requirements:

- **Status:** pending → in_progress → completed → skipped.
- **Drag & Drop:** Reorder tasks in the UI.
- **Smart Suggestions:** "You have 2 weak topics in Physics. Want to add DPP for Topic X?"

Acceptance Criteria:

ID	Criterion
AC-PL03	Task status transitions properly tracked.
AC-PL04	Task reordering persists.

FR-PLAN-03: Conflict with Gating

Priority: HIGH | **Status:** Approved

Description: Planner must respect gating constraints.

Requirements:

- Auto-generated tasks cannot suggest locked content.
- **Logic:** If Topic X is locked (Gate not passed), planner suggests "Complete Lecture Quiz for Topic X" instead of "Attempt DPP".

Acceptance Criteria:

ID	Criterion
AC-PL05	Auto-generated tasks only suggest unlocked content.
AC-PL06	Locked content suggestions properly redirected to prerequisite.

6.11 Notification System

FR-NOT-01: Multi-Channel Notifications

Priority: MEDIUM | **Status:** Approved

Description: System must notify users through multiple channels.

Channels:

Channel	Description
In-App	Bell icon with unread count
Email	Transactional (Welcome, Approval, Password Reset, Storage Alert, Verification)
Web Push	Reminders ("Your Daily Plan is ready", "Streak about to break", "Exam starting soon")

Triggers:

- User approved.
- New task generated.
- Streak milestone (7 days).
- Storage warning (Admin only).
- **Exam Reminder:** "Your scheduled test starts in 15 mins."

- **Email Verification:** "Verify your email address to activate your account."

Acceptance Criteria:

ID	Criterion
AC-N01	All notification channels functional.
AC-N02	Notifications delivered within 5 minutes of trigger.

FR-NOT-02: Notification Preferences

Priority: LOW | **Status:** Approved

Description: Users must control notification preferences.

Requirements:

- **User Control – Toggleable:** Daily Plan reminders; Streak reminders; Marketing/News (if added later).
- **System Critical Alerts** (e.g., Account Suspended, Email Verification Required) cannot be disabled.

Acceptance Criteria:

ID	Criterion
AC-N03	User preferences respected for all toggleable notifications.
AC-N04	Critical alerts always delivered.

6.12 Security & Access Control

FR-SEC-01: Gate Bypass Protection (Server-Side Enforcement)

Priority: HIGH | **Status:** Approved

Description: Real security must be server-side, not UI-only.

Requirements:

- Any API endpoint requesting gated content must call gate-checker Edge Function.
- If prerequisites not met, API returns 403 Forbidden with error code GATE_PREREQ_NOT_MET.
- **Prevention:** Prevents users from manipulating frontend state or guessing API endpoints.

Acceptance Criteria:

ID	Criterion
AC-S01	Gate bypass attempts return 403.
AC-S02	API validation independent of client state.

FR-SEC-02: Rate Limiting & Abuse Prevention

Priority: HIGH | Status: Approved

Description: Prevent spam and brute-force attacks.

Requirements:

Action	Limit
Quiz Attempts	Max 5 attempts per minute per user
PDF Uploads	Max 5 uploads per hour per user
Registration	Max 10 registrations per IP per hour
Exam Submission	Max 1 submission per exam instance
Password Change	Max 3 attempts per hour
Profile Photo Upload	Max 5 uploads per hour

Implementation: Supabase Edge Functions with Redis (or in-memory cache).

Acceptance Criteria:

ID	Criterion
AC-S03	Rate limits enforced consistently.
AC-S04	Exceeding limits returns 429 Too Many Requests.

FR-SEC-03: Data Isolation (Row Level Security)

Priority: HIGH | **Status:** Approved

Description: Strict data isolation between users.

Requirements:

- student can ONLY see their own user_notes, pomodoro_sessions, study_tasks, exam_attempts, profile_photos.
- student can ONLY see platform_content if is_approved = true AND content is unlocked.
- admin can see all platform_content but NO other user's personal data.
- super_admin can see ALL data (for support/debugging).

Acceptance Criteria:

ID	Criterion
AC-S05	RLS prevents access to other users' notes.
AC-S06	Role-based access properly enforced.

6.13 Enhanced Registration & Authentication System

FR-AUTH-01: Enhanced Registration Fields

Priority: HIGH | **Status:** Approved

Description: Registration form must collect specific user details with strict validation.

Requirements:

- **Full Name:** Required, alphanumeric with spaces, min 2 characters.
- **Mobile Number:** Required, must be valid Indian mobile number.

- **Format:** +91 prefix mandatory.
- **Length:** Exactly 10 digits after prefix.
- **Validation:** Regex `^(\+91)[\s-]?[6-9]\d{9}$`.
- **Error Handling:** Clear message for invalid format, non-Indian numbers, or non-numeric characters.
- **Email Address:** Required, valid email format.
- **Password:** Required, strong password policy.
 - Min 8 characters.
 - Must contain uppercase, lowercase, number, and special character.
 - Show/Hide toggle button available.
- **Confirm Password:** Required, must match Password field.
 - Show/Hide toggle button available.

Acceptance Criteria:

ID	Criterion
AC-AUTH01	Registration fails if mobile number is not valid Indian format (+91 + 10 digits).
AC-AUTH02	Registration fails if passwords do not match.
AC-AUTH03	Registration fails if password does not meet strength criteria.
AC-AUTH04	Show/Hide toggles function correctly for password fields.
AC-AUTH05	Validation errors are displayed inline and clearly.

FR-AUTH-02: Email Verification Workflow

Priority: HIGH | **Status:** Approved

Description: Users must verify their email address before accessing the platform.

Requirements:

- **Trigger:** Upon successful registration, a verification email is sent immediately via Resend.

- **Content:** Email contains a unique, time-limited verification link.
- **Login Restriction:** Users with email_verified = false cannot log in or access any content.
- **Resend Functionality:** "Resend Verification Email" button available on login screen for unverified users.
- **Success Handling:** Upon clicking link, user is redirected to login with success message.
- **Error Handling:** Expired or invalid links show appropriate error message.

Acceptance Criteria:

ID	Criterion
AC-AUTH06	Unverified users receive 403 Forbidden on login attempt.
AC-AUTH07	Verification email is sent within 30 seconds of registration.
AC-AUTH08	Resend button is rate-limited (max 3 times/hour).
AC-AUTH09	Verified users can log in immediately.

FR-AUTH-03: Password Management

Priority: HIGH | Status: Approved

Description: Secure password change functionality for authenticated users.

Requirements:

- **Access:** Available via Authorized User Profile Menu.
- **Fields:**
 - Current Password (Required).
 - New Password (Required, strong validation).
 - Confirm New Password (Required, must match).
- **Toggle:** Show/Hide buttons for all password fields.
- **Validation:**
 - Current password must be correct.
 - New password must meet strength criteria.
 - New password cannot be same as current.
- **Security:** Requires re-authentication (current password) before update.

Acceptance Criteria:

ID	Criterion
AC-AUTH10	Password change fails if current password is incorrect.
AC-AUTH11	Password change fails if new password is weak.
AC-AUTH12	Password change fails if new password matches old password.
AC-AUTH13	Show/Hide toggles function correctly in password change modal.

6.14 User Profile Management

FR-PROF-01: Profile Photo Management

Priority: MEDIUM | **Status:** Approved

Description: Users can manage their profile photo using **Backblaze B2**.

Requirements:

- **Upload:** Select image from device (JPG, PNG, WEBP).
- **Change:** Replace existing photo with new one.
- **Remove:** Delete profile photo (reverts to default avatar).
- **Processing:**
 - Client-side compression to max 500KB.
 - Resize to max 500x500 pixels.
 - Validate file type and size (< 2MB before compression).
- **Storage:** Uploaded to users/{uid}/profile/photo.jpg in Backblaze B2.
- **Security:** Only the user can upload/change their own photo.

Acceptance Criteria:

ID	Criterion
AC-PROF01	Upload fails if file type is not JPG/PNG/WEBP.
AC-PROF02	Upload fails if file size > 2MB (before compression).
AC-PROF03	Image is compressed and resized before upload.
AC-PROF04	Old photo is replaced/deleted from B2 on update.
AC-PROF05	Remove action deletes photo from B2 and reverts to default.

FR-PROF-02: Profile Information Display

Priority: MEDIUM | **Status:** Approved

Description: Profile menu displays user details.

Requirements:

- **Display Fields:**
 - Profile Photo (Circular avatar).
 - Full Name.
 - Mobile Number (Masked: +91 XXXXX XXXXX).
 - Email Address (Verified status indicator).
- **Editability:** Name and Mobile can be edited (subject to validation). Email is read-only (change requires re-verification).

Acceptance Criteria:

ID	Criterion
AC-PROF06	Profile menu displays all required fields correctly.
AC-PROF07	Mobile number is masked for privacy.
AC-PROF08	Verified email shows a checkmark icon.

7. External Interface Requirements

7.1 User Interfaces

7.1.1 Learning Path Dashboard

Description: Main navigation interface showing hierarchical progress.

Components:

- **Visual Tree:** Subject → Chapter → Topic hierarchy.
- **States:**
 -  **Locked:** Grayed out, tooltip "Complete previous step".

-  **Unlocked:** Active, clickable.
-  **Completed:** Green checkmark.
- **Next Action Card:** "Complete Lecture Quiz to unlock DPP".

7.1.2 Topic Detail View

Description: Detailed view for individual topic content.

Components:

Step	Description
Step 1	Watch Lecture (Telegram Link)
Step 2	Lecture Quiz (Locked until Step 1 clicked)
Step 3	DPP (Locked until Quiz Passed)
Step 4	PYQs (Locked until DPP Completed)
Step 5	Topic Test (Locked until PYQs Completed)
Step 6	"Upload Notes" Button (Visible only if Chapter Test Passed)

7.1.3 Dashboard (Revised)

Description: User home screen with productivity tools.

Components:

- **Top Bar:**
 - Focus Streak:  5 Days.
 - Today's Focus: 45 mins / 2h Goal.
 - **Profile Avatar:** Clickable to open Profile Menu.
- **Main Section:**
 - **Today's Plan** (Auto + Manual):
 - ☐ Complete Lecture Quiz: Physics Ch2 (Locked ) → Suggestion: Watch Lecture.
 - ☐ Revise Chemistry Formulas (Auto-generated).
 - ☐ Practice DPP: Math Ch1 (Unlocked ).
 - **Quick Actions:** [Start Pomodoro] (Big Button), [Add Manual Task].

- **Bottom Section:** Performance Trends: Internal vs. External Test Scores.
- **Profile Menu (Dropdown):**
 - **Header:** Profile Photo (Circular), Full Name, Mobile (Masked), Email (with Verified Badge).
 - **Actions:**
 - "Edit Profile" (Name/Mobile).
 - "Change Photo" (Upload/Change/Remove).
 - "Change Password" (Opens Modal).
 - "Logout".

7.1.4 Exam Interface (New)

Description: Full-screen, distraction-free exam environment.

Components:

- **Header:** Timer, User Name, "Pause", "Submit".
- **Main Area:** Question text, Options (Radio buttons), "Mark for Review" checkbox.
- **Sidebar/Grid:** Question numbers with color-coded status (Green/Red/Yellow/Gray).
- **Footer:** "Save & Next", "Previous", "Clear Response".
- **Mobile Adaptation:** Stacked layout, swipe gestures.

7.1.5 Result & Analysis View (New)

Description: Post-exam detailed breakdown.

Components:

- **Summary Card:** Score, Accuracy, Time, Rank.
- **Charts:** Pie chart (Correct/Incorrect), Bar chart (Topic-wise).
- **Tabs:**
 - **Answer Sheet:** List of questions with user vs. correct answer.
 - **Solutions:** Text explanations.
 - **Videos:** QR codes and embedded players.
 - **Insights:** "Weak in Thermodynamics", "Slow in Calculus".

7.1.6 Pomodoro Widget

Description: Floating timer component.

Components:

- **Floating Button:** Always accessible.
- **Modal:** Large timer, pause/resume, skip break.
- **Context:** "Currently studying: Physics Ch2" (if on a topic page).

7.1.7 Test Tracker View

Description: Performance analytics dashboard.

Components:

- **Table:** Date | Exam Name | Score | Percentile | Trend (▲/▼).
- **Chart:** Line graph overlaying Internal Test Avg vs. External Test Avg.

7.1.8 Registration & Authentication Screens

Description: Secure onboarding and login flows.

Components:

- **Registration Form:**
 - Full Name Input.
 - Mobile Number Input (with +91 prefix, regex validation).
 - Email Input.
 - Password Input (with strength meter, Show/Hide toggle).
 - Confirm Password Input (with Show/Hide toggle).
 - "Register" Button (Disabled until all validations pass).
 - "Already have an account? Login" Link.
- **Login Form:**
 - Email Input.
 - Password Input (Show/Hide toggle).
 - "Login" Button.
 - "Forgot Password?" Link.
 - **Unverified State:** If email not verified, show "Verify Email" banner with "Resend" button.
- **Password Change Modal:**
 - Current Password Input (Show/Hide).
 - New Password Input (Show/Hide, strength meter).
 - Confirm New Password Input (Show/Hide).
 - "Update Password" Button.
- **Profile Edit Modal:**

- Profile Photo Preview (with Upload/Change/Remove buttons).
- Full Name Input.
- Mobile Number Input (Validated).
- Email Display (Read-only with Verified Badge).
- "Save Changes" Button.

7.2 Hardware Interfaces

Description: No specialized hardware required.

Supported Devices:

- Desktop computers (Windows, macOS, Linux).
- Tablets (iPad, Android tablets).
- Smartphones (iOS 14+, Android 10+).
- Minimum screen resolution: 320px width.

Input Methods:

- Keyboard and mouse (desktop).
- Touch input (mobile/tablet).
- Voice entry (iOS app widget – future enhancement).
- Camera (for QR scanning and profile photo upload).

7.3 Software Interfaces

7.3.1 Supabase Platform (Own Credentials)

Interface Type: REST API + WebSocket

Component	Purpose	Protocol
Authentication	User login/signup, Email Verification	REST API
PostgreSQL	Data persistence	REST API + RPC
Storage	Small assets (< 100KB), Profile Photos	REST API
Edge Functions	Business logic, Mobile Validation	HTTPS
Realtime	Live updates	WebSocket

Authentication Method: JWT tokens via Supabase Auth.

Credential Structure:

Variable	Description
SUPABASE_URL	Project URL (e.g., https://xxxx.supabase.co)
SUPABASE_ANON_KEY	Public anonymous key (client-safe)
SUPABASE_SERVICE_ROLE_KEY	Admin key (Edge Functions only, never exposed to client)

Security Requirements:

- All credentials stored in Supabase Edge Function secrets (not in client code).
- SUPABASE_SERVICE_ROLE_KEY used only for privileged operations (gate checks, quota validation, email verification logic).
- RLS policies enforced on all tables to prevent unauthorized access.
- Email verification status checked on every login attempt.

7.3.2 Backblaze B2 Storage (Own Credentials)

Interface Type: S3-Compatible REST API

Operation	Endpoint	Purpose
Upload	PUT /file/{bucket}/{key}	Store PDF files and Profile Photos
Download	GET /file/{bucket}/{key}	Retrieve PDF files and Profile Photos
Delete	POST /b2_delete_file_version	Remove PDF files and Profile Photos
List	GET /b2_list_file_names	Enumerate user files

Authentication: Backblaze B2 Application Key (server-side only via Edge Functions).

Credential Structure:

Variable	Description	--- --- ---	B2_ACCOUNT_ID	Backblaze
account ID		B2_APPLICATION_KEY_ID	Key ID for authentication	

B2_APPLICATION_KEY | Secret key (never exposed to client) | |
B2_BUCKET_NAME | Target bucket for PDF and Photo storage |

Security Requirements:

- B2 Application Key stored in Supabase Edge Function secrets.
- Client never receives B2 credentials; only receives time-limited Signed URLs.
- All B2 buckets configured as **Private** with SSE-B2 encryption.
- CDN layer (Cloudflare) caches content to reduce B2 egress.
- Profile photos stored in users/{uid}/profile/ directory.

7.3.3 Telegram

Interface Type: Deep Links (No API dependency)

Function	Mechanism	Purpose
Lecture Delivery	tg://resolve?domain={username}&post={message_id}	Open video content
Click Tracking	Client-side event logging	Analytics only

«**Note:** Telegram API NOT used for completion tracking. »

7.3.4 YouTube

Interface Type: Embedded Player / URL Redirect

Function	Mechanism	Purpose
Video Solutions	https://youtube.com/watch?v={id}	QR redirect target
Embed	iframe	In-app video player (optional)

7.3.5 Resend

Interface Type: REST API

Email Type	Trigger	Content
Welcome	User registration	Account setup instructions
Verification	User registration	Email verification link
Approval	Admin approves user	Access granted notification
Password Reset	User request	Secure reset link
Storage Alert	B2 > 8GB	Admin notification
Resend Verification	User clicks "Resend"	New verification link

7.4 Communication Interfaces

7.4.1 Network Protocols

Protocol	Port	Usage
HTTPS	443	All client-server communication
WebSocket	443	Real-time updates (Supabase Realtime)

7.4.2 Data Formats

Format	Usage
JSON	API request/response bodies
JWT	Authentication tokens
PNG/JPG/WEBP	UI images and Profile Photos
PDF	User notes and documents
SVG	QR Codes

7.4.3 API Endpoints

Gate Validation:

```
POST /api/gate/check
Request:  { target_id: UUID, type: string }
Response: { allowed: boolean, next_step: string, reason?: string }
```

Exam Start:

```
POST /api/exam/start
Request:  { exam_id: UUID }
Response: { exam_token: string, start_time: timestamp, duration: int }
```

Exam Submit:

```
POST /api/exam/submit
Request:  { exam_token: string, answers: [{question_id, selected_option}],
Response: { score: int, accuracy: float, result_id: UUID }
```

Get Results:

```
GET /api/exam/results/{result_id}
Response: { score, accuracy, question_wise: [], solutions: [] }
```

QR Generation:

```
POST /api/qr/generate
Request:  { url: string, level: string (subject/chapter/topic/question) }
Response: { qr_code_svg: string, qr_code_png_url: string }
```

PDF Upload (Backblaze B2):

```
POST /api/b2/upload-url
Request:  { chapter_id: UUID, file_name: string, file_size: number }
Response: { upload_url: string, b2_key: string, expires_in: 900 }
```

PDF Download (Backblaze B2):

```
POST /api/b2/download-url
Request:  { note_id: UUID }
Response: { download_url: string, expires_in: 900 }
```


Profile Photo Upload (Backblaze B2):

```
POST /api/b2/profile-upload-url
Request:  { file_name: string, file_size: number }
Response: { upload_url: string, b2_key: string, expires_in: 900 }
```

Profile Photo Download (Backblaze B2):

```
POST /api/b2/profile-download-url
Request:  { user_id: UUID }
Response: { download_url: string, expires_in: 900 }
```

Storage Quota Check:

```
GET /api/b2/quota
Request:  { user_id: UUID }
Response: { user_usage_bytes: number, user_limit_bytes: number, global_usag
```

Email Verification:

```
POST /api/auth/verify-email
Request:  { token: string }
Response: { verified: boolean, message: string }
```

Resend Verification Email:

```
POST /api/auth/resend-verification
Request:  { email: string }
Response: { success: boolean, message: string }
```

Password Change:

```
POST /api/auth/change-password
Request:  { current_password: string, new_password: string }
Response: { success: boolean, message: string }
```

8. Non-Functional Requirements (Enhanced)

This section defines the critical quality attributes of the system. All requirements are measurable, testable, and implementation-ready.

8.1 Reliability

The ability of the system to perform its required functions under stated conditions for a specified period.

ID	Requirement	Metric	Implementation
NFR-REL-01 Fault Tolerance	System must continue to operate without data loss in transient network failures or client-side crashes	100% of user exam answers recoverable after browser crash or network interruption	IndexedDB local caching of exam state; retry logic with exponential backoff
NFR-REL-02 Error Handling	All system errors caught, logged, and presented with clear non-technical message	Zero unhandled exceptions reaching UI; 100% of errors logged to Sentry	Global error boundary in React; try-catch in all Edge Functions; user-friendly error pages
NFR-REL-03 Data Integrity	Data must remain consistent and accurate across all transactions	0 data corruption incidents; 100% ACID compliance	DB transactions for multi-step writes; constraints (FK, unique); regular integrity checks
NFR-REL-04 Recovery Time Objective (RTO)	Time to restore service must be minimized	RTO < 15 minutes for critical services (Auth, Exam Submission)	Automated health checks; pre-configured rollback scripts; redundant edge function deployments

8.2 Performance (Speed)

ID	Requirement	Metric	Implementation
NFR-PERF-01 Response Time	API responses and page loads fast enough for engagement	API: < 200ms p95; Gate Check: < 50ms; FCP: < 2s on 3G; Timer Sync: < 1s drift	Edge functions close to users; DB indexing; CDN caching; lazy loading
NFR-PERF-02 Load Handling	Handle expected concurrency without degradation	20 users < 1% errors; 500 < 2%; 2,000+ < 5%; Peak: 3x normal traffic	Horizontal scaling; PgBouncer; read replicas; CDN
NFR-PERF-03 Scalability	Architecture must allow seamless expansion	10x capacity increase with < 2 hours manual intervention	IaC (Terraform); stateless Edge Functions; sharding readiness; auto-scaling
NFR-PERF-04 Resource Utilization	Run efficiently on client devices	< 100MB RAM/tab; < 30% avg CPU; < 5MB initial page load (cached)	Code splitting; image optimization; Zustand; SW caching

8.3 Usability

ID	Requirement	Metric	Implementation
NFR-USE-01 Accessibility	Adhere to international standards	WCAG 2.1 AA compliance	Semantic HTML; ARIA labels; keyboard nav; high-contrast mode
NFR-USE-02 Learnability	New users perform core tasks with minimal instruction	90% of new users complete onboarding + first exam within 10 minutes unassisted	Intuitive UI; tooltips; guided tour; clear errors

ID	Requirement	Metric	Implementation
NFR-USE-03 UI Clarity & Consistency	Visual elements consistent across pages/devices	100% adherence to Design System	Component library (Tailwind + Framer Motion); design tokens; visual regression testing
NFR-USE-04 Mobile Responsiveness	Optimal experience on all screen sizes	Zero layout breaks 320px–2560px; Touch targets > 44x44px	Mobile-first CSS; flexible grids; responsive images
NFR-USE-05 Exam Usability	Intuitive and distraction-free exam interface	Users navigate (start/pause/submit) within 30 seconds	Minimalist design; clear status indicators; keyboard shortcuts; large touch targets
NFR-USE-06 Registration Usability	Clear validation feedback for mobile/email/password	< 5 seconds to understand validation errors	Inline error messages, real-time validation, clear regex hints

8.4 Security

ID	Requirement	Metric	Implementation
NFR-SEC-01 Authentication	Verify identity securely before access	100% access requires valid JWT; password ≥ 8 chars, mixed case, numbers	Supabase Auth; JWT (1h access, 7d refresh); rate limiting; account

ID	Requirement	Metric	Implementation
			lockout after 10 failures
NFR-SEC-02 Authorization	Users access only permitted data/functions	0 unauthorized access incidents; 100% queries via RLS	RLS on all tables; role-based middleware; least privilege
NFR-SEC-03 Data Protection	Encrypted in transit and at rest	100% TLS 1.3 in transit; 100% encrypted at rest	HTTPS everywhere; Supabase encryption; B2 SSE-B2 encryption; secure key management
NFR-SEC-04 Input Validation & Sanitization	All inputs validated to prevent injection	0 SQL injection or XSS vulnerabilities in pen testing	Parameterized queries; output encoding; CSP; file type/size validation; Mobile regex validation
NFR-SEC-05 Exam Integrity	Prevent cheating and ensure fair assessment	Detect tab-switching; prevent answer manipulation; server-side timer	Visibility API; server-side state validation; anti-tamper exam tokens
NFR-SEC-06 Credential Security	No credentials exposed to client	100% of secret keys stored in Edge Function secrets	Supabase Secrets Manager; B2 keys never in client code; Signed URLs only
NFR-SEC-07 Email Verification	Prevent unverified account access	100% of logins blocked until email_verified=true	Middleware check on login; RLS policies

8.5 Maintainability

ID	Requirement	Metric	Implementation
NFR-MAIN-01 Code Modularity	Logical, reusable, loosely coupled modules	> 90% coverage on critical paths; < 10% cyclomatic complexity in core	Modular architecture; unit/integration tests; linting
NFR-MAIN-02 Documentation	Code, APIs, processes well- documented	100% public APIs documented; up- to-date README & diagrams	Swagger/OpenAPI; JSDoc/TSDoc; Confluence/Notion
NFR-MAIN-03 Ease of Updates	Deployable with minimal downtime	Deploy < 5 mins; zero-downtime FE; hot-reload Edge Functions	CI/CD (GitHub Actions/Vercel); feature flags; blue- green
NFR-MAIN-04 Version Control	All changes tracked	100% code in Git; semantic versioning; clear commits	Git flow; PR reviews; automated changelog

8.6 Compatibility

ID	Requirement	Metric	Implementation
NFR-COMP-01 Browser Compatibility	Function on all major modern browsers	Full: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+; graceful degradation	Polyfills; cross- browser testing (BrowserStack); progressive enhancement
NFR-COMP-02 Platform Independence	Seamless across desktop, tablet, mobile	Identical functionality + sync across iOS, Android, Windows, macOS, Linux	Responsive design; platform-agnostic APIs; SW caching

ID	Requirement	Metric	Implementation
NFR-COMP-03 Device Adaptability	Adapt to screen sizes, resolutions, inputs	Touch, mouse, keyboard; portrait & landscape	Media queries; touch handlers; flexible grids
NFR-COMP-04 Network Adaptability	Function under varying conditions (3G to broadband)	Core features offline; graceful degradation; auto sync on reconnect	Offline-first; service workers; optimistic UI; retries

8.7 Availability

ID	Requirement	Metric	Implementation
NFR-AVAIL-01 Uptime Guarantee	Available during all operating hours	99.9% monthly ($\approx$ 43 min downtime); 99.99% during exam windows	Redundant infra; health monitoring; automatic failover; provider SLAs
NFR-AVAIL-02 Failure Recovery	Recover quickly from failures	MTTR < 1 hour for critical failures; automated alerts	Backups; DR plan; incident procedures; monitoring dashboards
NFR-AVAIL-03 Maintenance Windows	Communicate and minimize maintenance	< 30 min windows; 48-hour notice; zero data loss	Rolling updates; maintenance banners; off-peak scheduling

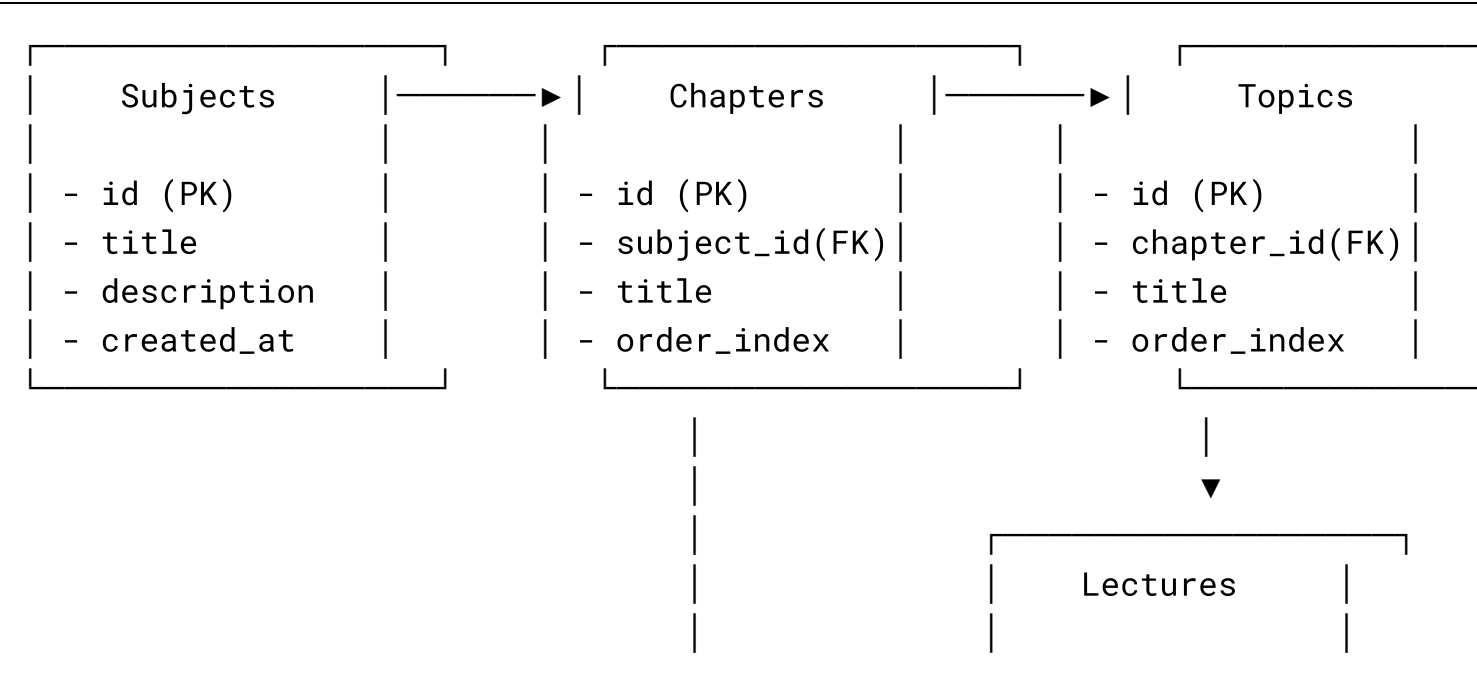
8.8 Flexibility (Scalability & Extensibility)

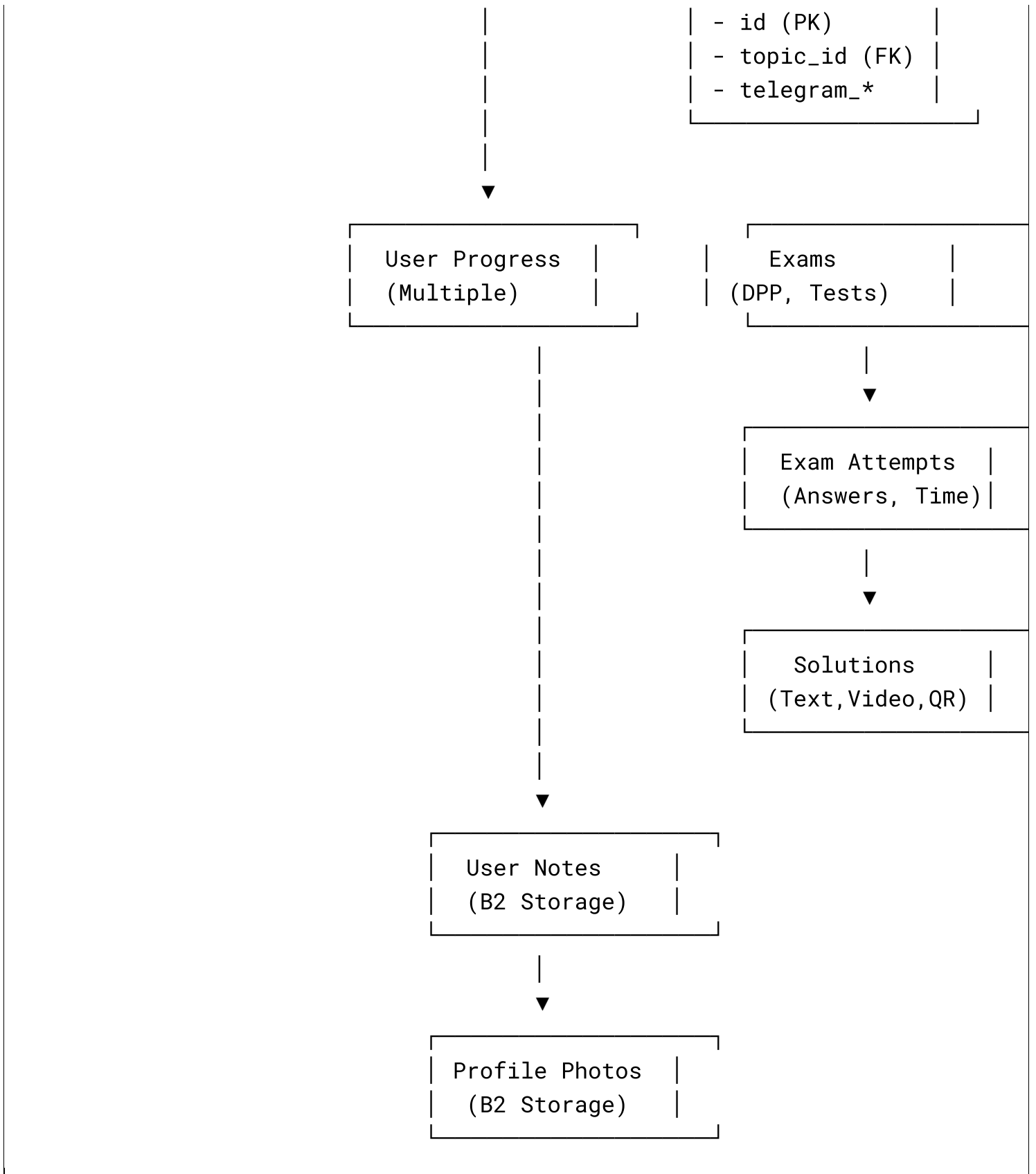
ID	Requirement	Metric	Implementation
NFR-FLEX-01 Scalability	Scale horizontally without architectural changes	Linear scaling; handle 10x growth	Stateless architecture; microservices-ready; sharding readiness; cloud-native

ID	Requirement	Metric	Implementation
NFR-FLEX-02 Extensibility	New features/modules added without disruption	New features < 20% impact on existing code; plugin architecture	Modular structure; API-first; abstraction layers; configuration-driven
NFR-FLEX-03 Configurability	System behavior configurable without code	100% of business rules configurable via admin UI / DB	system_config table; admin dashboard; environment variables
NFR-FLEX-04 Future-Proofing	Modern, supported tech stack	Use LTS versions; avoid deprecated libs	Regular updates; tech roadmap; community evaluation

9. Data/Database Design

9.1 Conceptual Data Model





9.2 Physical Database Schema

9.2.1 Core Hierarchy Tables

```
-- Subjects
CREATETABLE subjects (    id UUID PRIMARYKEYDEFAULT
gen_random_uuid(),    title TEXTNOTNULL,    description TEXT,
created_at TIMESTAMPTZ DEFAULTNOW());-- Chapters (Parent:
Subject)
CREATETABLE chapters (    id UUID PRIMARYKEYDEFAULT
gen_random_uuid(),    subject_id UUID REFERENCES
subjects(id)ONDELETECASCADE,    title TEXTNOTNULL,    order_index
INTNOTNULL,    created_at TIMESTAMPTZ DEFAULTNOW());-- Topics (Parent:
Chapter)
CREATETABLE topics (    id UUID PRIMARYKEYDEFAULT
gen_random_uuid(),    chapter_id UUID REFERENCES
chapters(id)ONDELETECASCADE,    title TEXTNOTNULL,    order_index
INTNOTNULL,    created_at TIMESTAMPTZ DEFAULTNOW());
```

9.2.2 Content & Gating Tables

```
-- Lectures (Telegram Links)
CREATETABLE lectures (    id UUID
PRIMARYKEYDEFAULT gen_random_uuid(),    topic_id UUID REFERENCES
topics(id)ONDELETECASCADE,    telegram_chat_id BIGINT,
telegram_message_id BIGINT,    is_active BOOLEANDEFAULTTRUE);-- Lecture
Clicks (Tracking only)
CREATETABLE lecture_clicks (    id UUID
PRIMARYKEYDEFAULT gen_random_uuid(),    user_id UUID REFERENCES
profiles(id),    lecture_id UUID REFERENCES lectures(id),    clicked_at
TIMESTAMPTZ DEFAULTNOW());-- Quizzes (All test types including DPPs and
Tests)
CREATETABLE quizzes (    id UUID PRIMARYKEYDEFAULT
gen_random_uuid(),    topic_id UUID REFERENCES topics(id),-- Nullable for
higher level tests    chapter_id UUID REFERENCES chapters(id),
subject_id UUID REFERENCES subjects(id),    title
TEXTNOTNULL,typeTEXTNOTNULLCHECK(typeIN('lecture_quiz','dpp','topic_test'
,'chapter_test','subject_test','grand_test')),    passing_score
INTDEFAULT60,-- Percentage    max_attempts INTDEFAULT3,
duration_minutes INT,-- For timed exams    negative_marking
DECIMALDEFAULT0,-- Negative marks per wrong answer    is_active
BOOLEANDEFAULTTRUE,    created_at TIMESTAMPTZ DEFAULTNOW());-- Quiz
Questions
CREATETABLE quiz_questions (    id UUID PRIMARYKEYDEFAULT
```

```

gen_random_uuid(),    quiz_id UUID REFERENCES quizzes(id)ONDELETECASCADE,
question_text TEXTNOTNULL,    options JSONB NOTNULL,--
{"A":"opt1","B":"opt2",...}    correct_answer TEXTNOTNULL,-- "A"
explanation TEXT,-- Text solution    difficulty INT,-- 1-5    order_index
INT,    video_solution_url TEXT,-- YouTube URL for specific question
qr_code_url TEXT-- Generated QR URL);-- DPPs & PYQs are now a type of
'quizzes' with type='dpp' / 'pyq'

```

9.2.3 User Progress & Gating Tables

```

-- User Topic Progress (Tracks Gating State)CREATETABLE
user_topic_progress (    user_id UUID REFERENCES profiles(id),
topic_id UUID REFERENCES topics(id),    lecture_clicked
BOOLEANDEFAULTFALSE,    lecture_quiz_passed BOOLEANDEFAULTFALSE,
lecture_quiz_score INT,    dpp_completed BOOLEANDEFAULTFALSE,
dpp_score INT,    pyqs_completed BOOLEANDEFAULTFALSE,    pyqs_score INT,
topic_test_passed BOOLEANDEFAULTFALSE,    topic_test_score INT,
topic_complete BOOLEANDEFAULTFALSE,    updated_at TIMESTAMPTZ
DEFAULTNOW(),PRIMARYKEY(user_id, topic_id));-- User Chapter
ProgressCREATETABLE user_chapter_progress (    user_id UUID REFERENCES
profiles(id),    chapter_id UUID REFERENCES chapters(id),
all_topics_complete BOOLEANDEFAULTFALSE,    chapter_test_attempted
BOOLEANDEFAULTFALSE,    chapter_test_passed BOOLEANDEFAULTFALSE,
pdf_upload_unlocked BOOLEANDEFAULTFALSE,-- Derived from Chapter Test
updated_at TIMESTAMPTZ DEFAULTNOW(),PRIMARYKEY(user_id, chapter_id));--
User Subject ProgressCREATETABLE user_subject_progress (    user_id UUID
REFERENCES profiles(id),    subject_id UUID REFERENCES subjects(id),
all_chapters_complete BOOLEANDEFAULTFALSE,    subject_test_attempted
BOOLEANDEFAULTFALSE,    subject_test_passed BOOLEANDEFAULTFALSE,
updated_at TIMESTAMPTZ DEFAULTNOW(),PRIMARYKEY(user_id, subject_id));--
User Attempts (Unified for Quizzes, DPP, PYQ, Tests)CREATETABLE
user_attempts (    id UUID PRIMARYKEYDEFAULT gen_random_uuid(),
user_id UUID REFERENCES profiles(id),    quiz_id UUID REFERENCES
quizzes(id),    content_id UUID NOTNULL,-- Redundant but explicit for
clarity    content_type TEXTNOTNULLCHECK(content_type
IN('quiz', 'dpp', 'pyq', 'test'))),    score INT,    total_marks INT,

```

```

accuracy DECIMAL,      time_taken_ms INT,      negative_marks_applied
DECIMAL,statusTEXTDEFAULT'submitted'CHECK(statusIN('in_progress','paused'
,'submitted','expired','abandoned'))),      submitted_at TIMESTAMPTZ,
started_at TIMESTAMPTZ,      created_at TIMESTAMPTZ DEFAULTNOW(),
is_correct_summary JSONB                                -- Snapshot of
correct/incorrect status);-- Detailed User Answers (One row per question
per attempt)CREATETABLE user_answers (      id UUID PRIMARYKEYDEFAULT
gen_random_uuid(),      attempt_id UUID REFERENCES
user_attempts(id)ONDELETECASCADE,      question_id UUID REFERENCES
quiz_questions(id),      selected_option TEXT,      is_correct BOOLEAN,
time_spent_ms INT,      is_marked_for_review BOOLEANDEFAULTFALSE,
created_at TIMESTAMPTZ DEFAULTNOW());

```

9.2.4 PDF Storage (Gated - Backblaze B2)

```

-- User Notes (PDFs) - Gated by Chapter Completion, Stored in
B2CREATETABLE user_notes (      id UUID PRIMARYKEYDEFAULT
gen_random_uuid(),      user_id UUID REFERENCES profiles(id)NOTNULL,
chapter_id UUID REFERENCES chapters(id)NOTNULL,-- Scoped to Chapter
title TEXTNOTNULL,      content_jsonb JSONB,-- Rich text      b2_storage_path
TEXT,-- B2 key path (replaces r2_key)      b2_file_id TEXT,-- B2 file
version ID      pdf_mime_type TEXT,      pdf_size_bytes BIGINT,
pdf_text_index TEXT,-- Extracted text for search      file_hash TEXT,--
SHA-256 hash for deduplication      tags TEXT[],      created_at TIMESTAMPTZ
DEFAULTNOW(),      updated_at TIMESTAMPTZ DEFAULTNOW(),      search_vector
TSVECTOR GENERATED ALWAYS AS(to_tsvector('english',coalesce(title,'')||'
'||coalesce(pdf_text_index,'')))) STORED
);-- Trigger to enforce Chapter Completion before PDF
UploadCREATEORREPLACEFUNCTION
check_chapter_completion_for_upload()RETURNSTRIGGERAS $$
DECLARE      chapter_status BOOLEAN;BEGINSELECT pdf_upload_unlocked INTO
chapter_status
FROM user_chapter_progress
WHERE user_id = NEW.user_id AND chapter_id = NEW.chapter_id;IFNOT
chapter_status THEN      RAISE EXCEPTION 'Access Denied: You must
complete the Chapter Test to upload notes for this chapter.';ENDIF;RETURN

```

```

NEW;END;$$ LANGUAGE plpgsql;CREATE TRIGGER trg_check_chapter_upload
BEFORE INSERT OR UPDATE ON user_notes
FOR EACH ROW EXECUTE FUNCTION check_chapter_completion_for_upload();--
Index for full-text search CREATE INDEX idx_user_notes_search ON user_notes
USING GIN(search_vector);

```

9.2.5 Profile Photos (Backblaze B2)

```

-- User Profile Photos - Stored in B2
CREATE TABLE user_profile_photos (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES profiles(id) NOT NULL UNIQUE,
  b2_storage_path TEXT NOT NULL, -- B2 key path:
  users/{uid}/profile/photo.jpg
  b2_file_id TEXT, -- B2 file version ID
  mime_type TEXT,
  size_bytes BIGINT,
  created_at TIMESTAMPTZ
  DEFAULT NOW(),
  updated_at TIMESTAMPTZ DEFAULT NOW());
-- RLS Policy:
Users can only see their own photo
CREATE POLICY "Users can view own
profile photo" ON user_profile_photos
FOR SELECT USING (auth.uid() = user_id);
CREATE POLICY "Users can insert own
profile photo" ON user_profile_photos
FOR INSERT WITH CHECK (auth.uid() = user_id);
CREATE POLICY "Users can update
own profile photo" ON user_profile_photos
FOR UPDATE USING (auth.uid() = user_id);
CREATE POLICY "Users can delete own
profile photo" ON user_profile_photos
FOR DELETE USING (auth.uid() = user_id);

```

9.2.6 Pomodoro Sessions

```

CREATE TABLE pomodoro_sessions (
  id UUID PRIMARY KEY DEFAULT
  gen_random_uuid(),
  user_id UUID REFERENCES profiles(id) NOT NULL,
  topic_id UUID REFERENCES topics(id), -- Optional context
  duration_seconds INT NOT NULL,
  break_duration_seconds INT DEFAULT 300,
  start_time TIMESTAMPTZ NOT NULL,
  end_time TIMESTAMPTZ NOT NULL,
  is_completed BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ

```

```
DEFAULTNOW());-- Index for streak calculationCREATEINDEX  
idx_pomo_user_time ON pomodoro_sessions(user_id, start_time);
```

9.2.7 Study Tasks (Planner)

```
CREATETABLE study_tasks (    id UUID PRIMARYKEYDEFAULT gen_random_uuid(),  
user_id UUID REFERENCES profiles(id)NOTNULL,    title TEXTNOTNULL,  
description TEXT,-- Target Link (Polymorphic)    target_type  
TEXTCHECK(target_type  
IN('platform_subtopic','personal_topic','free_text')),    target_id  
UUID,-- Nullable if free_text-- Task  
MetadatastatusTEXTDEFAULT'pending'CHECK(statusIN('pending','in_progress',  
'completed','skipped')),    priority TEXTDEFAULT'medium'CHECK(priority  
IN('low','medium','high')),    due_date TIMESTAMPTZ,-- Source    source  
TEXTDEFAULT'manual'CHECK(source IN('auto','manual')),    created_at  
TIMESTAMPTZ DEFAULTNOW(),    completed_at TIMESTAMPTZ,CONSTRAINT  
fk_target CHECK((target_type ='free_text'AND target_id  
ISNULL)OR(target_type ='platform_subtopic'ANDEXISTS(SELECT1FROM topics  
WHERE id = target_id))OR(target_type  
='personal_topic'ANDEXISTS(SELECT1FROM personal_topics WHERE id =  
target_id)))));CREATEINDEX idx_study_tasks_user_status ON  
study_tasks(user_id,status);CREATEINDEX idx_study_tasks_due_date    ON  
study_tasks(due_date);
```

9.2.8 External Test Logs

```
CREATETABLE external_tests (    id UUID PRIMARYKEYDEFAULT  
gen_random_uuid(),    user_id UUID REFERENCES profiles(id)NOTNULL,  
exam_name TEXTNOTNULL,    exam_date DATENOTNULL,    score_obtained  
NUMERICNOTNULL,    total_marks NUMERICNOTNULL,    percentile NUMERIC,  
rank INT,    notes TEXT,    created_at TIMESTAMPTZ
```

```
DEFAULTNOW());CREATEINDEX idx_external_tests_user_date ON
external_tests(user_id, exam_date);
```

9.2.9 System Configuration & Audit

```
-- System ConfigurationCREATETABLE system_config (keyTEXTPRIMARYKEY,value
JSONB NOTNULL,    updated_at TIMESTAMPTZ DEFAULTNOW(),    updated_by UUID
REFERENCES profiles(id));-- Initial Configuration ValuesINSERTINTO
system_config (key,value)VALUES('uploads_paused','false'),
('global_storage_limit_gb','9'),('per_user_storage_limit_mb','500'),
('lecture_quiz_passing_score','60'),('topic_test_passing_score','70'),
('max_quiz_attempts','3'),('max_exam_pauses','2'),
('exam_timeout_warning_mins','5'),('mobile_validation_regex','^(\+91)
[\s-]?[6-9]\d{9}$'),('password_min_length','8'),
('require_email_verification','true');-- Audit LogsCREATETABLE audit_logs
(    id UUID PRIMARYKEYDEFAULT gen_random_uuid(),    actor_id UUID
REFERENCES profiles(id),actionTEXTNOTNULL,    target_type TEXT,
target_id UUID,    old_value JSONB,    new_value JSONB,    ip_address
INET,    created_at TIMESTAMPTZ DEFAULTNOW());CREATEINDEX
idx_audit_logs_actor    ON audit_logs(actor_id);CREATEINDEX
idx_audit_logs_created ON audit_logs(created_at);
```

9.2.10 Profiles Extension (Enhanced)

```
-- Extend Supabase auth.users tableCREATE EXTENSION IFNOTEXISTS"uuid-
ossp";-- Add custom fields to auth.users (or create a public.profiles
table linked to it)-- Note: In Supabase, it is best practice to create a
public.profiles table that mirrors auth.usersCREATETABLEpublic.profiles (
id UUID REFERENCES auth.users(id)ONDELETECASCADEPRIMARYKEY,    full_name
TEXTNOTNULL,    mobile_number TEXTNOTNULL,-- Stores +91XXXXXXXXXX
email TEXTNOTNULL,    role TEXTDEFAULT'student',    is_approved
BOOLEANDEFAULTFALSE,statusTEXTDEFAULT'pending_approval',
```

```

email_verified BOOLEAN DEFAULT FALSE,    created_at TIMESTAMPTZ
DEFAULT NOW(),    updated_at TIMESTAMPTZ DEFAULT NOW());-- Trigger to
create profile on
signup CREATE OR REPLACE FUNCTION public.handle_new_user() RETURN TRIGGER AS $$
BEGIN INSERT INTO public.profiles (id, full_name, mobile_number, email,
role, is_approved, status, email_verified) VALUES (
NEW.id, COALESCE(NEW.raw_user_meta_data->>'full_name', 'User'), COALESCE(NEW.raw_user_meta_data->>'mobile_number', ''),
NEW.email, CASE WHEN (SELECT COUNT(*) FROM public.profiles) = 0 THEN 'super_admin' ELSE 'student' END, CASE WHEN (SELECT COUNT(*) FROM public.profiles) = 0 THEN TRUE ELSE FALSE END, CASE WHEN (SELECT COUNT(*) FROM public.profiles) = 0 THEN 'approved' ELSE 'pending_approval' END, FALSE -- Email not verified yet); RETURN NEW; END; $$
LANGUAGE plpgsql SECURITY DEFINER; CREATE TRIGGER on_auth_user_created
AFTER INSERT ON auth.users
FOR EACH ROW EXECUTE FUNCTION public.handle_new_user();-- Indexes for
performance CREATE INDEX idx_profiles_mobile
ON public.profiles (mobile_number); CREATE INDEX idx_profiles_email_verified
ON public.profiles (email_verified); CREATE INDEX idx_profiles_role
ON public.profiles (role); CREATE INDEX idx_profiles_approval
ON public.profiles (is_approved);

```

9.3 Data Dictionary

Table	Column	Type	Description	Constraints
subjects	id	UUID	Unique identifier	PRIMARY KEY
subjects	title	TEXT	Subject name	NOT NULL
chapters	subject_id	UUID	Parent subject	FOREIGN KEY

Table	Column	Type	Description	Constraints
topics	chapter_id	UUID	Parent chapter	FOREIGN KEY
quizzes	type	TEXT	Type of assessment	CHECK constraint
user_attempts	status	TEXT	Attempt state	CHECK constraint
user_answers	selected_option	TEXT	User's choice	–
user_notes	b2_storage_path	TEXT	B2 storage key	UNIQUE
user_notes	file_hash	TEXT	SHA-256 hash	For deduplication
user_profile_photos	b2_storage_path	TEXT	B2 photo path	UNIQUE
system_config	key	TEXT	Configuration key	PRIMARY KEY
profiles	mobile_number	TEXT	Indian mobile (+91)	NOT NULL, Regex validated
profiles	email_verified	BOOLEAN	Email verification status	Default FALSE
profiles	full_name	TEXT	User's full name	NOT NULL

9.4 Data Retention & Archival

NFR-DATA-01: Archive Policy

Priority: MEDIUM

Requirements:

- user_attempts: Archive records older than 12 months to attempts_archive.

- **Trigger:** Monthly cron job (storage-monitor Edge Function).
- **Benefit:** Keeps main DB fast and within Supabase Free Tier row limits.

Implementation:

```
CREATE TABLE attempts_archive (LIKE user_attempts INCLUDING ALL);--
Monthly archival query
INSERT INTO attempts_archive
SELECT * FROM user_attempts
WHERE created_at < NOW()-INTERVAL '12 months';
DELETE FROM user_attempts
WHERE created_at < NOW()-INTERVAL '12 months';
```

NFR-DATA-02: Backup Strategy

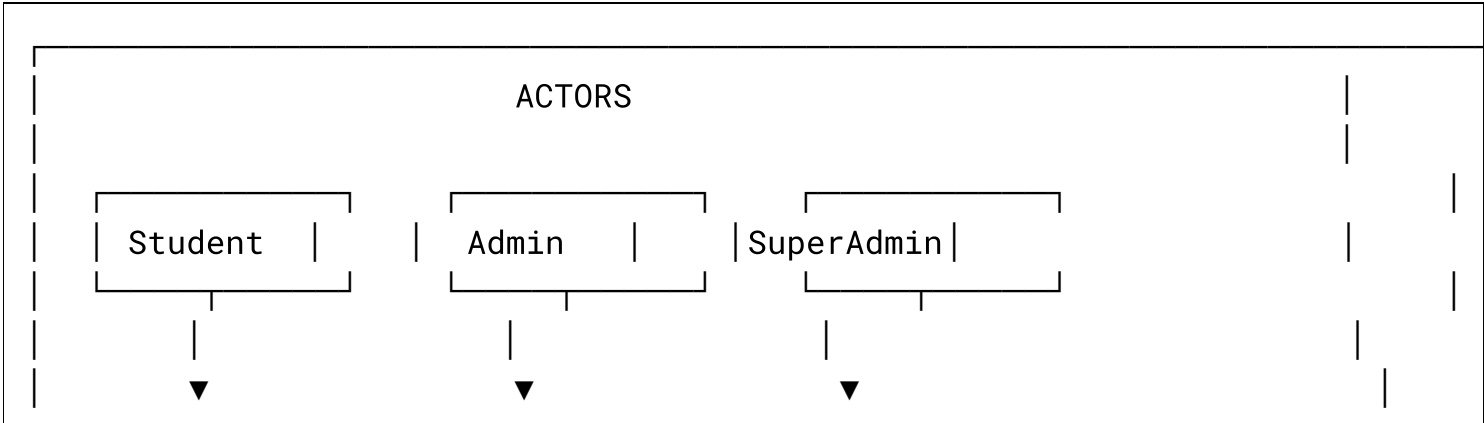
Priority: HIGH

Requirements:

- **Database:** Enable Supabase automated daily backups (Free Tier).
- **B2 Storage:** Enable Versioning on the B2 bucket (Keep last 3 versions) for both PDFs and Profile Photos.
- **Export:** super_admin can export all user data (JSON) for GDPR compliance.

10. Use Cases & System Models

10.1 Use Case Diagram Overview



SYSTEM BOUNDARY				
[Browse Content]	[Take Exams]	[Upload Notes]		
[Track Progress]	[View Solutions]	[Manage Tasks]		
[Scan QR Codes]	[Use Pomodoro]	[Log External Tests]		
[Register]	[Verify Email]	[Manage Profile]		
[Change Password]	[Upload Photo]	[Edit Details]		
[Create Content]	[Configure Gates]	[Approve Users]		
[Bulk Import]	[Manage Exams]	[Reset Progress]		
[Upload Solutions]	[Generate QR Codes]			

10.2 Detailed Use Cases

UC-01: Student Progress Through Gated Content

Field	Value
Actor	Student
Preconditions	User is authenticated, email verified, and approved
Postconditions	User has completed topic and can access next content

Main Success Scenario:

- Student navigates to Learning Path Dashboard.
- System displays unlocked topic (Lecture step).
- Student clicks "Watch Lecture" → Telegram deep link opens.
- System logs lecture click event.
- Student returns and takes Lecture Quiz.
- Student passes quiz (score ≥ passing threshold).
- System updates lecture_quiz_passed = true.
- DPP becomes unlocked.
- Student completes DPP with sufficient score.
- System updates dpp_completed = true.
- PYQs become unlocked.

2. Process continues through Topic Test.
3. Topic marked complete.
4. When all topics complete, Chapter Test unlocks.
5. After Chapter Test, PDF Upload unlocks.

Alternative Flows:

- **A1: Quiz Failure** – Student scores below threshold; System allows retry (up to max_attempts); If attempts exhausted, block until admin reset or time delay.
- **A2: Direct URL Access Attempt** – Student tries to access DPP via direct URL; System calls gate-checker Edge Function; Gate check fails → Returns 403 Forbidden.

Special Requirements:

- **SR-01:** Gate check must complete in < 50ms.
- **SR-02:** All state changes must be atomic.

UC-02: Admin Content Creation

Field	Value
Actor	Admin
Preconditions	User has admin role
Postconditions	New content available to students

Main Success Scenario:

1. Admin accesses Content Management Dashboard.
2. Admin creates new Subject (or selects existing).
3. Admin creates Chapters within Subject.
4. Admin creates Topics within Chapters.
5. Admin adds Lecture (Telegram deep link).
6. Admin creates Exam (DPP/Test) with questions.
7. Admin sets passing score, duration, and negative marking.
8. Admin adds text solutions and YouTube URLs for video solutions.
9. System auto-generates QR codes for videos.
0. Admin saves content.
1. System validates hierarchy integrity.

2. Content becomes available to students (when approved).

Alternative Flows:

- **A1: Bulk Import** – Admin uploads CSV/JSON; System validates hierarchy; System imports in batch; Errors reported for invalid entries.

Special Requirements:

- **SR-03:** Bulk import must handle 1000+ questions.
- **SR-04:** Validation must catch orphaned content.

UC-03: Super Admin User Management

Field	Value
Actor	Super Admin
Preconditions	User has super_admin role
Postconditions	User status updated

Main Success Scenario:

1. Super Admin accesses User Management Dashboard.
2. System displays pending approval queue.
3. Super Admin reviews user profile.
4. Super Admin clicks "Approve" or "Reject".
5. System updates user status.
6. If approved, user gains content access.
7. If rejected, user receives notification.

Alternative Flows:

- **A1: Role Promotion** – Super Admin promotes student to admin; System logs audit entry; New role takes effect immediately.
- **A2: Emergency Recovery** – Only super_admin account compromised; Founder uses recovery code via secure Edge Function; New super_admin role assigned.

Special Requirements:

- **SR-05:** Sole super_admin cannot demote themselves.

- **SR-06:** All role changes logged in audit_logs.

UC-04: PDF Upload (Gated - Backblaze B2)

Field	Value
Actor	Student
Preconditions	Chapter test completed, user approved, email verified
Postconditions	PDF stored in B2 and searchable

Main Success Scenario:

1. Student navigates to completed chapter.
2. System displays "Upload Notes" button (unlocked).
3. Student selects PDF file (< 10MB).
4. System validates file type and size.
5. System checks user storage quota (< 500MB).
6. System checks global storage quota (< 9GB).
7. System generates presigned URL for B2 via Edge Function.
8. Student uploads file directly to B2.
9. System extracts text via pdf.js.
0. System indexes text for search.
1. System creates record in user_notes.
2. Upload confirmation displayed.

Alternative Flows:

- **A1: Chapter Not Complete** – Student attempts upload on incomplete chapter; System shows "Locked" state with motivational message.
- **A2: Storage Quota Exceeded** – User reaches 500MB limit; System returns 501 error with cleanup suggestion.
- **A3: Global Storage Full** – B2 usage > 9GB; System returns 429 error, uploads paused.
- **A4: Duplicate File Detected** – File hash matches existing; System prompts user to confirm overwrite or skip.

Special Requirements:

- **SR-07:** Text extraction must complete in < 30 seconds.
- **SR-08:** Deleted notes must remove B2 files.

- **SR-09:** B2 Signed URLs must expire after 15 minutes.

UC-05: Exam Attempt (Enhanced)

Field	Value
Actor	Student
Preconditions	User is authenticated, email verified, exam is unlocked and active
Postconditions	Exam submitted, results generated

Main Success Scenario:

1. Student clicks "Start Exam" on a Topic Test.
2. System validates prerequisites and starts timer.
3. Exam interface loads with Question Grid and Timer.
4. Student reads Question 1, selects Option B.
5. Student clicks "Mark for Review" (Yellow flag).
6. Student navigates to Question 5 via Grid.
7. Student pauses exam (Timer stops).
8. Student resumes exam later.
9. Student answers remaining questions.
0. Timer reaches 00:00 → Auto-submit triggered.
1. System calculates score, applies negative marking.
2. Result page displays score, accuracy, and insights.
3. Student clicks "View Answer Sheet".
4. Student scans QR code for Question 3 video solution.

Alternative Flows:

- **A1: Network Disconnection** – Internet drops during exam; System saves answers locally; User reconnects; answers sync automatically.
- **A2: Browser Crash** – Browser closes unexpectedly; User reopens app; "Resume Exam" prompt appears; User resumes from last saved state.
- **A3: Missing Video** – QR code scanned, but video URL is broken; System shows "Video unavailable" but displays text solution.

Special Requirements:

- **SR-09:** Timer must sync server-client to prevent cheating.

- **SR-10:** Auto-submit must capture all unsaved answers.
- **SR-11:** QR codes must generate valid URLs.

UC-06: Pomodoro Session Management

Field	Value
Actor	Student
Preconditions	User is authenticated, email verified
Postconditions	Session logged, streak updated

Main Success Scenario:

1. Student clicks "Start Pomodoro" button.
2. System initializes timer (default 25 min).
3. Student optionally links session to current Topic.
4. Timer runs in foreground/background.
5. Student completes session (timer finishes).
6. System plays completion chime.
7. System logs session to pomodoro_sessions.
8. System updates "Focus Streak" counter.
9. Dashboard reflects new stats.

Alternative Flows:

- **A1: Session Aborted** – Student pauses or stops early; Session not logged (or logged as incomplete); Streak not incremented.
- **A2: Offline Mode** – Student starts timer without internet; Timer runs locally (IndexedDB); On reconnect, session syncs to server.

Special Requirements:

- **SR-12:** Timer must persist across page refreshes.
- **SR-13:** Background execution on supported browsers.

UC-07: Smart Planner Task Generation

Field	Value
Actor	System (Automated)

Field	Value
Preconditions	Cron job triggers at 00:00 UTC
Postconditions	Daily tasks created for users

Main Success Scenario:

1. study-planner Edge Function triggers.
2. System fetches v_weak_topics (accuracy < 50%).
3. System fetches due_revision_tasks (spaced repetition).
4. System filters out locked topics (gate check).
5. System generates 3-5 suggested tasks.
6. Tasks inserted into study_tasks with source='auto'.
7. Push notification sent to users.

Alternative Flows:

- **A1: No Weak Topics** – System defaults to revision tasks; if none, creates generic "Review" task.
- **A2: Locked Content Detected** – System suggests "Complete Lecture Quiz" instead of "Attempt DPP".

Special Requirements:

- **SR-14:** Generation must complete within 5 minutes.
- **SR-15:** Tasks must respect gating logic.

UC-08: External Test Logging

Field	Value
Actor	Student
Preconditions	User is authenticated, email verified
Postconditions	External test recorded in analytics

Main Success Scenario:

1. Student navigates to Test Tracker.
2. Student clicks "Log External Test".
3. Student enters exam details (Name, Date, Score, Rank).

4. Student submits form.
5. System validates data ranges.
6. System inserts record into external_tests.
7. Analytics chart updates immediately.

Alternative Flows:

- **A1: Invalid Data** – Score > Total Marks → System rejects with validation error.

Special Requirements:

- **SR-16:** Chart must render within 1 second.

UC-09: User Registration with Enhanced Validation

Field	Value
Actor	Prospective Student
Preconditions	None
Postconditions	User account created, email verification sent, pending approval

Main Success Scenario:

1. User navigates to Registration page.
2. User enters Full Name, Mobile (+91XXXXXXXXXX), Email, Password, Confirm Password.
3. System validates Mobile format (Regex), Password strength, and matching passwords.
4. User clicks "Register".
5. System creates Supabase Auth user and profiles record.
6. System assigns role (Super Admin if first, else Student).
7. System sends verification email via Resend.
8. User receives "Check your email" message.

Alternative Flows:

- **A1: Invalid Mobile Number** – System detects non-Indian format or wrong length; Shows inline error "Please enter a valid Indian mobile number (+91...)".

- **A2: Weak Password** – System detects missing criteria; Shows specific missing requirements.
- **A3: Email Already Exists** – System detects duplicate; Shows "Account already exists. Please login."

Special Requirements:

- **SR-17:** Mobile validation must be client-side and server-side.
- **SR-18:** Email verification link must be unique and time-limited.

UC-10: Email Verification

Field	Value
Actor	Student
Preconditions	User registered but email not verified
Postconditions	User email verified, login enabled

Main Success Scenario:

1. User clicks verification link in email.
2. System validates token.
3. System updates email_verified = true in profiles.
4. User is redirected to Login page with success message.
5. User logs in successfully.

Alternative Flows:

- **A1: Expired Token** – System shows "Link expired. Please request a new one."
- **A2: Invalid Token** – System shows "Invalid verification link."
- **A3: Already Verified** – System shows "Email already verified. Please login."

Special Requirements:

- **SR-19:** Login attempts must fail if email_verified = false.

UC-11: Profile Photo Management

Field	Value
Actor	Authenticated Student
Preconditions	User is logged in, email verified
Postconditions	Profile photo updated in B2

Main Success Scenario:

1. User opens Profile Menu.
2. User clicks "Change Photo".
3. User selects image file (JPG/PNG/WEBP, < 2MB).
4. System compresses and resizes image client-side.
5. System requests Signed Upload URL from Edge Function.
6. System uploads image to B2 (users/{uid}/profile/photo.jpg).
7. System updates user_profile_photos table.
8. System deletes old photo from B2 (if exists).
9. UI updates with new photo.

Alternative Flows:

- **A1: Invalid File Type** – System rejects non-image files.
- **A2: File Too Large** – System rejects files > 2MB before compression.
- **A3: Remove Photo** – User clicks "Remove"; System deletes from B2 and DB; UI reverts to default avatar.

Special Requirements:

- **SR-20:** Image compression must maintain quality while reducing size.
- **SR-21:** Old photos must be deleted to save B2 storage.

UC-12: Password Change

Field	Value
Actor	Authenticated Student
Preconditions	User is logged in, email verified
Postconditions	Password updated securely

Main Success Scenario:

1. User opens Profile Menu → "Change Password".
2. User enters Current Password, New Password, Confirm New Password.
3. System validates Current Password.
4. System validates New Password strength.
5. System updates password in Supabase Auth.
6. System logs out all other sessions (optional security feature).
7. User sees success message.

Alternative Flows:

- **A1: Wrong Current Password** – System rejects update; Shows "Current password is incorrect."
- **A2: Weak New Password** – System rejects update; Shows strength requirements.
- **A3: Same as Old** – System rejects update; Shows "New password cannot be same as current."

Special Requirements:

- **SR-22:** Password change requires re-authentication.

11. Error Handling & Edge Cases

11.1 Error Classification

Category	Description	Examples
Validation Errors	Input fails business rules	Score < passing threshold, File too large, Invalid mobile format, Weak password
Authorization Errors	User lacks permissions	Accessing locked exam, Viewing others' answers, Unverified login attempt
System Errors	Infrastructure failures	Database timeout, B2 connection failure, QR generation fail, Email

Category	Description	Examples
		service down
Network Errors	Connectivity issues	Timeout, DNS failure, 3G dropouts during exam, Upload interruption
Data Integrity Errors	Corrupted or inconsistent data	Orphaned records, broken foreign keys, mismatched answers

11.2 Specific Error Scenarios & Handling

EC-01: Gate Bypass Attempt

- **Scenario:** User tries to access DPP via direct URL without passing Lecture Quiz.
- **Detection:** API endpoint calls gate-checker Edge Function.
- **Response:**

Field	Value
HTTP Status	403 Forbidden
Error Code	GATE_PREREQ_NOT_MET
Message	"Prerequisite not met: Complete Lecture Quiz first."
UI Action	Redirect to previous unlocked step with tooltip

EC-02: Failed Quiz with Exhausted Attempts

- **Scenario:** User fails Lecture Quiz 3 times (max_attempts reached).
- **Detection:** user_attempts count checked against max_attempts.
- **Response:**

Field	Value
HTTP Status	403 Forbidden
Error Code	MAX_ATTEMPTS_EXCEEDED

Field	Value
Message	"Maximum attempts reached. Please wait 24 hours or contact admin."
UI Action	Disable quiz button, show countdown timer

EC-03: Storage Quota Exceeded (User)

- **Scenario:** User attempts to upload PDF exceeding 500MB personal limit.
- **Detection:** b2-upload Edge Function calculates current usage via B2 API.
- **Response:**

Field	Value
HTTP Status	413 Payload Too Large (or 501 Not Implemented)
Error Code	USER_STORAGE_LIMIT_EXCEEDED
Message	"You have reached your 500MB storage limit. Delete old notes to upload more."
UI Action	Show storage usage bar, link to "Manage Notes"

EC-04: Storage Quota Exceeded (Global)

- **Scenario:** B2 bucket usage exceeds 9GB.
- **Detection:** storage-monitor cron job or real-time check in b2-upload.
- **Response:**

Field	Value
HTTP Status	429 Too Many Requests
Error Code	GLOBAL_STORAGE_LIMIT_REACHED
Message	"System storage is full. Uploads are temporarily paused."
UI Action	Disable upload button globally, show admin alert banner

EC-05: Telegram Link Broken

- **Scenario:** Telegram message deleted or link invalid.
- **Detection:** Admin flag or user report.
- **Response:**
 - **UI Action:** Show "Link unavailable" message.
 - **Fallback:** Display "Contact Admin" button.
 - **Admin Action:** Update telegram_message_id or switch to local video.

EC-06: Race Condition in User Registration

- **Scenario:** Two users register simultaneously; both think they are the first.
- **Detection:** Database transaction with SERIALIZABLE isolation.
- **Response:**
 - **Mechanism:** on-user-created webhook + DB trigger fallback.
 - **Logic:** Atomic count check before role assignment.
 - **Outcome:** Only first user gets super_admin; second gets student + pending_approval.

EC-07: Offline Sync Conflict

- **Scenario:** User edits note offline, then syncs; another user edited same note (unlikely for personal notes, but possible for shared tasks).
- **Detection:** Version vector or timestamp comparison.
- **Response:**
 - **Strategy:** Last-write-wins (for personal notes).
 - **UI Action:** Show "Synced" status, log conflict in debug mode.
 - **Future Enhancement:** Merge conflicts UI (out of scope for v7.3).

EC-08: Database Connection Pool Exhaustion

- **Scenario:** 2,000 concurrent users hit Supabase connection limit.
- **Detection:** Connection pool metrics.
- **Response:**
 - **Mechanism:** Connection pooling (PgBouncer).
 - **Fallback:** Queue requests, return 503 Service Unavailable with retry-after header.
 - **Scaling:** Upgrade to paid Supabase tier.

EC-09: Malicious PDF Upload

- **Scenario:** User uploads PDF containing malware or prohibited content.
- **Detection:** File type validation + content scanning (future).
- **Response:**
 - **Immediate:** Block executable files (.exe, .js disguised as PDF).
 - **Post-upload:** super_admin can delete via moderation dashboard.
 - **Legal:** Report severe violations per Swiss law.

EC-10: Recovery Code Compromise

- **Scenario:** Founder's recovery code leaked.
- **Detection:** Unusual role change activity.
- **Response:**
 - **Mitigation:** Recovery code is single-use or time-limited.
 - **Action:** Rotate code immediately, audit logs reviewed.
 - **Prevention:** Store code in secure hardware vault (e.g., YubiKey).

EC-11: Exam Timer Desync

- **Scenario:** Client clock is faster/slower than server.
- **Detection:** Periodic server ping to adjust client timer.
- **Response:**
 - **Mechanism:** Server sends server_time in start_exam response.
 - **Logic:** Client calculates offset and adjusts countdown.
 - **Outcome:** Auto-submit triggers at correct absolute time.

EC-12: Missing Video Solution

- **Scenario:** Admin forgot to upload video URL for a question.
- **Detection:** Null check in quiz_questions.video_solution_url.
- **Response:**
 - **UI Action:** Hide QR code, show "Video solution coming soon" text.
 - **Fallback:** Ensure text explanation is always visible.
 - **Admin Alert:** Log warning in audit_logs for content gap.

EC-13: Exam Auto-Submit Failure

- **Scenario:** Network fails exactly at the moment of auto-submit timeout.
- **Detection:** Client-side timeout triggers submit request; server returns error.
- **Response:**

- **Mechanism:** Client retries submission with exponential backoff (up to 3 times).
- **Fallback:** If all retries fail, save full state to localStorage and mark attempt as "Pending Sync".
- **Recovery:** On next connection, a background worker attempts to push the "Pending Sync" attempt.
- **UI Action:** Show "Submission delayed due to network. Retrying..." toast.

EC-14: QR Code Generation Failure

- **Scenario:** YouTube API returns invalid URL or rate limit hit during QR generation.
- **Detection:** Edge Function qr-generator catches API error.
- **Response:**
 - **Fallback:** Generate a static QR pointing to a "Video Unavailable" landing page.
 - **Admin Notification:** Alert admin to manually update the video link.
 - **User Impact:** User sees QR but clicking leads to a helpful error message, not a broken link.

EC-15: Negative Marking Calculation Error

- **Scenario:** Floating point precision issues cause score to be slightly off (e.g., 49.999999 instead of 50).
- **Detection:** Unit tests and validation logic in submit_exam function.
- **Response:**
 - **Mechanism:** Use integer arithmetic (store marks in cents/smallest unit) or DECIMAL(10,2) with explicit rounding.
 - **Logic:** Round final score to 2 decimal places before display.
 - **UI Action:** Display rounded score; store precise value internally for tie-breakers if needed.

EC-16: Backblaze B2 Authentication Failure

- **Scenario:** B2 Application Key is invalid or expired.
- **Detection:** Edge Function receives 401/403 from B2 API.
- **Response:**
 - **Immediate:** Log critical error to Sentry and Admin Dashboard.

- **User Impact:** Return 500 Internal Server Error with generic message "Storage service temporarily unavailable."
- **Recovery:** Admin must rotate keys in Supabase Secrets Manager.

EC-17: B2 Signed URL Expiration

- **Scenario:** User takes longer than 15 minutes to upload a file.
- **Detection:** B2 returns 403 on PUT request.
- **Response:**
 - **UI Action:** Show "Upload link expired. Please try again."
 - **Recovery:** User clicks "Retry" to request a new Signed URL from Edge Function.

EC-18: CDN Cache Invalidation

- **Scenario:** User deletes a note, but CDN still serves the old file.
- **Detection:** User reports seeing deleted content.
- **Response:**
 - **Mechanism:** Edge Function triggers Cloudflare Cache Purge API for specific file path.
 - **Latency:** Cache purge takes < 60 seconds globally.
 - **Fallback:** If purge fails, serve 404 from B2 directly (bypassing CDN) for that specific request.

EC-19: Invalid Mobile Number Registration

- **Scenario:** User enters a non-Indian mobile number or invalid format.
- **Detection:** Client-side regex and server-side validation in Edge Function.
- **Response:**
 - **HTTP Status:** 400 Bad Request.
 - **Error Code:** INVALID_MOBILE_FORMAT.
 - **Message:** "Please enter a valid Indian mobile number starting with +91."
 - **UI Action:** Highlight input field with red border and show error message.

EC-20: Unverified Email Login Attempt

- **Scenario:** User tries to log in before verifying their email.
- **Detection:** Login middleware checks email_verified flag in profiles.

- **Response:**
 - **HTTP Status:** 403 Forbidden.
 - **Error Code:** EMAIL_NOT_VERIFIED.
 - **Message:** "Please verify your email address before logging in. Check your inbox or click 'Resend'."
 - **UI Action:** Show banner with "Resend Verification Email" button.

EC-21: Password Change Failure (Wrong Current Password)

- **Scenario:** User enters incorrect current password.
- **Detection:** Supabase Auth updateUser fails.
- **Response:**
 - **HTTP Status:** 400 Bad Request.
 - **Error Code:** WRONG_CURRENT_PASSWORD.
 - **Message:** "Current password is incorrect."
 - **UI Action:** Shake animation on password field, show error message.

EC-22: Profile Photo Upload Failure (Size Limit)

- **Scenario:** User uploads an image larger than 2MB.
- **Detection:** Client-side file size check.
- **Response:**
 - **HTTP Status:** 413 Payload Too Large.
 - **Error Code:** FILE_TOO_LARGE.
 - **Message:** "Image size must be less than 2MB."
 - **UI Action:** Show error message and disable upload button.

11.3 Error Logging & Monitoring

Logging Strategy:

Layer	Mechanism
Client-Side	Log errors to console (dev) + Sentry (prod)
Server-Side	Log to Supabase Edge Function logs + CloudWatch

Layer	Mechanism
Audit Trail	All security-relevant errors (gate bypass, role change, exam tampering, B2 auth fail, mobile validation fail, email verification fail) logged to audit_logs
Exam Specific	Log every pause, resume, and auto-submit event with timestamps

Alerting Thresholds:

Metric	Threshold	Action
Error Rate	> 5% of requests in 5 minutes	Slack alert
Latency	P95 > 200ms	Email alert
Storage	> 8.5GB	Email to super_admin
Uptime	< 99%	PagerDuty escalation
Exam Integrity	> 1% auto-submit failures	Immediate alert to DevOps
B2 Auth Failures	> 0 occurrences	Immediate alert to DevOps (Critical)
Mobile Validation Failures	> 10% of registrations	Alert to Product (Check regex)
Email Verification Failures	> 5% of registrations	Alert to Ops (Check Resend config)

12. Assumptions & Dependencies

12.1 Explicit Assumptions

- 1. **User Behavior:** Students will not systematically attempt to bypass gates; security is designed to prevent accidental or casual bypasses,

not determined hacking.

2. **Network Availability:** Users have intermittent but sufficient internet access for initial sync and periodic updates.
3. **Telegram Stability:** Telegram deep links remain functional; Telegram does not change its API or link format in a breaking way.
4. **Supabase Free Tier:** Supabase maintains current free tier limits (500MB storage, 500K rows, 2GB bandwidth) for the initial phase.
5. **Backblaze B2:** Backblaze maintains 10GB free storage with no hidden egress fees for the target usage pattern (mitigated by CDN).
6. **Device Capabilities:** Target devices support modern JavaScript (ES6+), IndexedDB, Service Workers, and image compression APIs.
7. **Legal Compliance:** Content uploaded by users complies with copyright laws; platform acts as a neutral host with moderation tools.
8. **Time Zones:** All timestamps stored in UTC; UI converts to user's local timezone.
9. **Language:** Primary interface language is English; other languages are translations of the English UI.
0. **Exam Syllabus:** The content hierarchy (Subjects/Chapters/Topics) aligns with standard competitive exam syllabi (e.g., JEE, NEET, GATE).
1. **Visual Preference:** Students prefer dark interfaces for night-time study sessions.
2. **Video Availability:** YouTube videos for solutions are hosted on public or unlisted channels and remain accessible.
3. **QR Scanning:** Users have access to a camera-enabled device or can use the in-app scanner.
4. **Network Latency:** Network latency between user and edge functions is $< 100\text{ms}$ for 95% of users.
5. **CDN Efficiency:** Cloudflare CDN successfully caches B2 content, reducing egress costs to near zero for repeated views.
6. **Mobile Validity:** Users possess valid Indian mobile numbers (+91) and can receive SMS/OTP if required in future phases (though email is primary verification now).
7. **Email Deliverability:** Resend service delivers verification emails reliably ($>95\%$ success rate) to major providers (Gmail, Outlook, Yahoo).
8. **Image Compression:** Client-side image compression maintains sufficient visual quality for profile avatars while reducing file size.

12.2 Dependencies

Dependency	Type	Criticality	Mitigation if Failed
Supabase	Infrastructure	High	Migrate to self-hosted PostgreSQL + Auth provider
Backblaze B2	Storage	High	Switch to AWS S3 or Cloudflare R2 (cost increase)
Cloudflare CDN	Optimization	Medium	Direct B2 access (increased egress cost)
Telegram	Content Delivery	Medium	Host videos on YouTube/Vimeo (requires paid plan for ad-free)
Vercel	Hosting	Medium	Migrate to Netlify or AWS Amplify
Resend	Email	Medium	Switch to SendGrid, AWS SES, or Postmark
Node.js v18+	Dev Env	Low	Upgrade/downgrade runtime version
Git	Version Control	Low	Use alternative VCS (Mercurial, SVN)
YouTube	Video Hosting	Medium	Host videos on self-hosted server (increases storage cost)
QR Library	Client Lib	Low	Switch to alternative QR generation library
Sentry	Monitoring	Low	Switch to alternative error tracking (e.g., LogRocket)
Image Compression Lib	Client Lib	Low	Switch to browser-image-compression or similar

12.3 Out of Scope (Future Releases)

- 1. **Native Mobile Apps:** iOS/Android native wrappers (PWA only for v7.3).

2. **AI Content Generation:** Automatic question generation or personalized learning paths.
 3. **Real-Time Collaboration:** Multi-user editing of notes or shared study sessions.
 4. **Payment Gateway:** Subscription billing (handled externally or manual for now).
 5. **Advanced Analytics:** Predictive modeling for exam performance (beyond current insights).
 6. **Voice Commands:** Voice-controlled navigation (currently text-only).
 7. **Social Features:** Leaderboards, peer discussion forums.
 8. **Proctoring:** Advanced AI proctoring (eye tracking, face detection) – currently only basic tab-switch warnings.
 9. **SMS Verification:** Mobile number verification via SMS OTP (currently only format validation).
 0. **Multi-Language Support:** UI localization beyond English.
-

13. Appendices

13.1 Glossary

Term	Definition
DPP	Daily Practice Problems: Short problem sets for daily practice.
PYQ	Previous Year Questions: Past exam questions for revision.
PWA	Progressive Web App: Web application with native-like features.
RLS	Row-Level Security: Database policy restricting row access.
B2	Backblaze B2: Object storage service with S3-compatible API.
CDN	Content Delivery Network: Cloudflare layer caching B2 content.

Term	Definition
Edge Function	Serverless function running at the network edge (Deno runtime).
Gate	Prerequisite check that blocks access to next content level.
Streak	Consecutive days of meeting a study goal (e.g., 4 Pomodoro sessions).
Super Admin	Highest privilege user; system bootstrap owner.
Gated Upload	PDF upload feature unlocked only after chapter completion.
Dark Mode	Primary UI theme optimized for low-light study environments.
Auto-Submit	Automatic exam submission when timer expires.
Mark for Review	Flagging a question to revisit later during an exam.
Negative Marking	Deduction of marks for incorrect answers.
QR Code	Quick Response code linking to video solutions.
MTTR	Mean Time To Recovery: Average time to restore service after failure.
SLA	Service Level Agreement: Contractual commitment for uptime/performance.
Signed URL	Time-limited URL granting temporary access to private B2 objects.
Email Verified	Boolean flag indicating user has confirmed ownership of email address.
Mobile Validated	Boolean flag (implied) indicating mobile number matches Indian regex.

13.2 List of Assumptions Made (Summary)

Category	Assumption
Technical	Supabase and Backblaze B2 free tiers remain stable and unchanged.
Behavioral	Users will not actively exploit system vulnerabilities.
Operational	Internet connectivity is sufficient for sync operations.
Content	Uploaded PDFs and images are legal and non-malicious.
Business	\$0/month cost model is sustainable for initial 20 users (via CDN).
Legal	Swiss law compliance covers all user-generated content moderation.
Design	PWA provides adequate experience compared to native apps.
Data	9GB storage is sufficient for 2,000 users with 500MB/user limit.
Security	JWT tokens, RLS, Signed URLs, and Email Verification provide sufficient protection.
Timeline	Development can be completed within the planned schedule using the specified stack.
Visual	Dark mode reduces eye strain and improves retention for long study sessions.
Media	YouTube videos remain accessible and unblocked in target regions.
Network	Latency remains within acceptable limits for real-time exam features.
Storage	Cloudflare CDN effectively reduces B2 egress to negligible levels.
Registration	Users possess valid Indian mobile numbers and can access email for verification.

13.3 Testing Strategy & Verification

Unit Testing:

- All Edge Functions and database triggers must have > 90% code coverage.
- Focus on logic for gating, scoring, timer synchronization, B2 Signed URL generation, mobile regex validation, and email verification flows.
- Test image compression logic and profile photo upload/download flows.

Integration Testing:

- Verify end-to-end flows: Registration (Mobile/Email/Pass) → Email Verification → Login → Content Access → Exam → Result.
- Test interactions between Frontend, Edge Functions, Supabase, and Backblaze B2.
- Verify B2 upload/download flows for both PDFs and Profile Photos with CDN caching.
- Test password change flow with re-authentication.

Load Testing:

- Simulate 2,000 concurrent users using k6.
- Verify API response times remain < 200ms (p95) and error rates < 5%.
- Verify B2 upload throughput and CDN cache hit ratios.
- Stress test registration endpoint to ensure mobile validation and email sending scale.

Security Testing:

- Penetration testing for SQL injection, XSS, CSRF, B2 credential leakage, and mobile number spoofing.
- Verify RLS policies prevent unauthorized data access (especially profile photos).
- Test exam integrity (tab switching, timer manipulation).
- Verify Signed URL expiration and revocation.
- Test email verification token validity and expiration.
- Verify password strength enforcement and re-authentication requirements.

Usability Testing:

- Conduct user trials with target demographic (16-25 years).
- Measure time-to-complete for key tasks (register, verify email, start exam, upload notes, change photo).
- Validate accessibility compliance (WCAG 2.1 AA).

- Test mobile number input experience on various devices (keyboard types, formatting).

13.4 Deployment & Maintenance Plan

Deployment Pipeline:

Component	Strategy
CI/CD	GitHub Actions for automated testing and deployment
Frontend	Deploy to Vercel on merge to main
Backend	Deploy Edge Functions on merge to main
Database	Migration scripts run automatically via Supabase CLI
Secrets	Managed via Supabase Dashboard (B2 keys, Resend keys, etc.)

Rollback Strategy:

- Automatic rollback if health checks fail post-deployment.
- Manual rollback capability via version tagging.

Monitoring:

Aspect	Tool
Uptime	Pingdom / UptimeRobot for endpoint monitoring
Errors	Sentry for real-time error tracking
Performance	Google Lighthouse for continuous performance scoring
Logs	Supabase Logs for database and Edge Function debugging
Storage	Backblaze B2 Dashboard for usage monitoring
Email	Resend Dashboard for delivery metrics

Maintenance Schedule:

Frequency	Activities
Weekly	Dependency updates and security patches
Monthly	Database archival and backup verification
Quarterly	Security audit and performance review
As Needed	B2 key rotation, Resend API key rotation

End of Document

«This Software Requirements Specification (SRS) is approved for production deployment. All stakeholders acknowledge the requirements defined herein. Any changes to this document must follow the formal change control process.**Infrastructure Note:** This system relies on **externally managed Supabase** and **Backblaze B2** using **own credentials**. No AI-platform built-in backends or storage are utilized.**Compliance Note:** All user registrations require valid Indian mobile numbers (+91) and verified email addresses. Profile photos and PDF notes are stored securely in Backblaze B2 with strict access controls. »